



Does economic policy uncertainty drive the dynamic spillover among traditional currencies and cryptocurrencies? The role of the COVID-19 pandemic

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ABSTRACT

The paper examines the dynamic spillover among traditional currencies and cryptocurrencies before and during the COVID-19 pandemic and investigates whether economic policy uncertainty (EPU) impacts this spillover. Based on the TVP-VAR approach, we find evidence of spillover effects among currencies, which increased widely during the pandemic. In addition, results suggest that almost all cryptocurrencies remain as “safe-haven” tools against market uncertainty during the COVID-19 period. Moreover, comparative analysis shows that the total connectedness for cryptocurrencies is lower than for traditional currencies during the crisis. Further analysis using quantile regression suggests that EPU exerts an impact on the total and the net spillovers with different degrees across currencies and this impact is affected by the health crisis. Our findings have important policy implications for policymakers, investors, and international traders.

1. Introduction

In the aftermath of the 2008 global financial crisis (GFC), economic policy uncertainty (EPU) has raised remarkably, suggesting an increasing influence on the financial markets, especially the foreign exchange markets (Krol, 2014; Arouri et al., 2016; Balcilar et al., 2016; Kido, 2016; Beckmann and Czudaj, 2017; Abid, 2020). In this regard, Baker, Bloom, and Davis (2016) invented new economic uncertainty indices using the information from newspaper articles for major economies, which have been popularly used in the recent literature. For instance, Krol (2014) and Chen, Du, and Hu (2020) argued that economic policy uncertainty adversely affects foreign currency values. Other studies argued that cryptocurrencies are also influenced by changes in EPU (Cheng and Yen, 2020; Demir et al., 2018; Paule-Vianez et al., 2020; Mokni, 2021; Elsayed et al., 2022). Further studies suggest that cryptocurrencies are not only used as means of exchange but also act as investment tools and safe havens to protect against economic uncertainty (Wu et al., 2019, 2021; Paule-Vianez et al., 2020; Hasan et al., 2021; Colon et al., 2021; Mokni et al., 2022).

As investors in foreign currency markets have various choices of investment opportunities in different currencies and/or currency

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futures contracts, any change in the value of a particular currency (or cryptocurrency) might transmit to the value and/or price volatility of other currencies, including cryptocurrencies, leading to a significant effect on portfolio investments, and diversification benefits. The higher connectedness among currencies within each currency system or among currencies in the other system is the higher information transmission among currencies. Therefore, studying the connectedness among both types of currencies can help identify how extreme volatilities and trends in one type of currency are transmitted within and across others. Such an analysis can also be useful for traders and policymakers to make suitable decisions and investors regarding portfolio diversification benefits and hedging opportunities, especially in periods of high uncertainty.

The degree of connectedness between traditional and digital currencies may also result from the differences in the characteristics of these two kinds of assets. Like traditional currencies, cryptocurrencies may function as a store of value, a medium of exchange, and a unit account. However, unlike traditional currencies, cryptocurrencies are not issued by central banks and can be transferred electronically between users without the involvement of intermediaries (Andrada-Félix et al., 2020; Bech and Garratt, 2017). They can also generate revenues similar to any regular goods and services and rely on the cryptographic integrity of their network, while traditional currencies depend on political and legal systems for value and legitimacy. Therefore, the difference in the characteristics of these two types of assets would lead to a difference in the degree of connectedness of currencies among each other and within every type of currency. If cryptocurrencies serve as a medium of exchange, there would be systematic traditional currency exposures in cryptocurrencies; when volatility connectedness among traditional currencies is high, this may generate extreme volatility transmission in cryptocurrencies, which in turn might be transmitted within cryptocurrencies and across traditional currencies as well.

In the last few years, we observed a significant increase in the cryptocurrency market, accompanied by a significant rise in the demand for traditional currencies, especially over the ongoing COVID-19 pandemic. However, the increase in the demand for both types of currencies has led to high variations in exchange rates and the value of cryptocurrencies. Both currencies are still considered valuable investment opportunities, either in speculation, hedging against risk, or portfolio diversification. In addition, cryptocurrencies are considered as a solution for unexpected economic and financial structure changes (Demir et al., 2018). Given the increasing importance of cryptocurrencies as an investment vehicle, modeling cryptocurrencies connectedness becomes necessary for making investment decisions and risk management (e.g., Katsiampa, 2019; Bouri et al., 2017; Bouri et al., 2021).

Most of the literature has reported evidence of connectedness among traditional currencies and/or cryptocurrencies (e.g., Kitamura, 2010; Baruník et al., 2017; Sehgal et al., 2017; Mokni and Ajmi, 2020). However, the literature has left several research gaps. The different studies have generally reached mixed results and inconclusive evidence of such connectedness. Several studies have focused mostly on the connectedness among one-one traditional currency and/or a few currencies, the world's major traditional currencies (Mai et al., 2018; Sehgal et al., 2017). Other studies only examined the connectedness among cryptocurrencies (Wei et al., 2020; Mai et al., 2018; Kumar and Anandaramo, 2019; Mensi et al., 2019; Aslan and Sensoy, 2020), and largely concentrated on Bitcoins and Ethereum. Moreover, another stream of studies examined the connectedness among traditional or cryptocurrencies with other financial assets, including commodities, bonds, oil prices, and stocks (Boako and Alagidede, 2017; Grobys, 2015; Malik and Umar, 2019; Antonakakis and Kizys, 2015; Hassen et al., 2022; Cao and Xie, 2022; Conlon et al., 2020). Given the above-uncovered research gaps, it can be seen that the dynamic connectedness among traditional currencies and cryptocurrencies is extremely understudied and still an attractive research venue.

Another issue which has been overlooked by the literature is the effect of EPU on the degree of connectedness among traditional currencies and cryptocurrencies. However, although the evidence of the effect of EPU on either type of currency is well established, there is still a need to re-examine such an effect due to the limited research effort on this topic. Mokni et al. (2020) have examined the effect of EPU on the connectedness among either type of currency but without taking into account the impact of the COVID-19 crisis. Although the effect of EPU has been examined by other studies (Chen et al., 2020; Huynh et al., 2020), they focused only on examining the effect on the connectedness among either cryptocurrencies or traditional currencies, not both. Therefore, it is of great interest to examine the effect of EPU on the connectedness among cryptocurrencies and fiat currencies during the COVID-19 outbreak.

The COVID-19 pandemic has caused a higher exchange rate risk exposure to many currencies, which has led to high demand for hard currencies as well as for cryptocurrencies, leading to an appreciation of these currencies. On the other hand, over the COVID-19 period, exchange rates and/or the global stock markets have significantly fluctuated with high downturns in the global economies (Narayan et al., 2020; Njindan, 2020; So et al., 2021; Samitas et al., 2022; Cheng et al., 2022; Guo et al., 2021). Given the negative impacts of the health crisis on the global economy and major global currencies as well as on cryptocurrencies, there has been a limited research investigation directed toward the relationship between EPU and the dynamic connectedness among traditional currencies and cryptocurrencies during the COVID-19 crisis. Therefore, exploring the impact of economic policy uncertainty (EPU) on the dynamic connectedness among currencies during the COVID-19 period is still an appealing research issue.

The main objectives of the paper are: 1) analyzing the dynamic connectedness among traditional currencies (the Euro (EUR), the British Pound (UK), the Japanese Yen (JPY), the Indian Rupee (IND), Swiss Franc (SFR), Korean Won (KOR), Chinese Yuan (CHIN), and the Canadian Dollar (CAN)), and cryptocurrencies (Bitcoin (BIT), Ethereum (ETH), Ripple (RIP) and Litecoin (LIT)) and 2) whether such connectedness, among both types of currencies, is affected by uncertainty factors under different levels of spillovers' distribution. To be specific, this paper contributes to the literature in two folds. First, it analyzes the dynamic connectedness and spillover effects among traditional currencies before and during the COVID-19 crisis. To perform the analysis, several currencies and cryptocurrencies are selected based on long-term trading history and according to their market capitalization. To measure the dynamic connectedness among currencies, the connectedness index and its variant are generated as proposed by Diebold and Yilmaz (2009, 2012, 2014). This index provides a clear insight into comparing the extent and the nature of interdependencies and spillover between both types of currencies. The variants of the index are based on the time-varying parameter vector autoregressive (TVP-VAR) framework. The TVP-VAR approach is based on a multivariate Kalman filter and is less sensitive to the presence of outliers (e.g., Antonakakis et al.,

2019; Gabauer and Gupta, 2018). This approach is more sophisticated than that based on the rolling-window approach. The latter approach outcomes some weaknesses related to the choice of the window size and the loss of information in calculating the dynamic measures of connectedness. We also apply the generalization of the connectedness among these two groups of currencies.

Second, our analysis includes the most recently updated data covering the ongoing COVID-19 crisis, which can detect the possible effect of this outbreak on the shock transmission between these two types of currencies. Our updated data sample covers the period between the beginning of 2017 and early October 2022. Third, our study analyzes the potential effect of the economic policy uncertainty index (EPU) on the dynamic connectedness among traditional currencies before and during the COVID-19 pandemic in different levels of dynamic spillovers. To perform this analysis, we implement a quantile regression model to investigate the possible effect of the EPU on the dynamic connectedness to evaluate such an effect for different levels of spillover indices' distribution. Therefore, this analysis enables us to assess the sensitivity of the information flows between currencies to those uncertainty factors under different phases of connectedness.

Our results are summarized as follows. We find strong evidence of dynamic spillover across currency markets, and the spillover level increased widely during the COVID-19 pandemic period. Specifically, almost all cryptocurrencies remain as "safe-haven" tools to hedge uncertainty, showing heterogeneous responses to the presence of the COVID-19 pandemic. The level of connectedness across traditional currencies is higher than the connectedness among cryptocurrencies during the crisis period. However, before the crisis, the connectedness for traditional currencies is lower than connectedness for cryptocurrencies. Based on the quantile regression analysis, we find that the total spillover index is negatively impacted by the economic policy uncertainty through the pre-COVID-19 and during COVID-19, indicating that total spillover among currencies is negatively and highly influenced by the rise in EPU in normal and high connectedness condition. However, EPU has different degrees of influence on the dynamic spillover across different conditions during the crisis. Although this impact fluctuates when the connectedness effects are low, our results show a decreasing pattern when the level of connectedness is medium or high. The total connectedness among all currencies becomes increasingly exposed to the negative impact of EPU in normal and high levels of connectedness.

The rest of the paper is organized as follows: The review of the related literature is discussed in Section 2. In Section 3, the methodology and data are explained. The empirical analysis is reported and discussed in Section 4, and the concluding remarks are addressed in Section 5.

2. Literature review

The literature examining the impact of EPU on the currency markets has been growing in recent years. However, the existing literature has provided conflicting conclusions. The recent studies can be categorized into two important strands of literature (see Appendix A for the summary of the literature review). The first strand of literature focused on volatility connectedness among global currency markets, mainly examining the volatility transmission among major global traded currencies. Using the M-GARCH model, Kitamura (2010) argued that volatility spillovers from the Euro drastically impact the Swiss franc and Japanese yen during the subprime global financial crisis (GFC) period. Baruník et al. (2017) found evidence of asymmetric volatility connectedness in the foreign exchange markets using a 2 N-dimensional VAR model. The most actively traded currencies (AUD, GBP, CAD, EUR, JPY, and CHF) also reported that volatility's negative spillovers dominate positive ones. Positive spillovers are correlated with the subprime crisis (GFC), while negative spillovers are generally caused by the sovereign debt crisis in Europe.

Using an empirical network model, Greenwood-Nimmo et al. (2016) examined the risk-return connectedness among the G10 currencies and reported evidence of time-varying connections between such currencies. During the periods of financial crises, volatility connectedness was increasing, and returns became more sensitive to risk measures. By using the VAR model, Salisu and Ayinde (2018) provided evidence of volatility transmission across the Naira and six most-traded currencies (the US Dollar, Euro, Pound Sterling, Yen, Swiss Franc, and the West African Unit of Account (WAUA)), indicating that the election process in Nigeria led to greater spillover effects on the Naira than the GFC.

In the context of the Asian markets, Shu et al. (2015) found evidence of transmission effects from the Chinese currency on the Asia-Pacific countries' currencies. The offshore currency exchange of the Chinese currency was found to exert more effect on Asian currencies than the onshore Chinese currency market due to China's monetary policy transmission to the region. Sehgal et al. (2017), using constant and time-varying Copula-GARCH models, found that currencies of the South Asian member countries, except for India and Nepal/Bhutan, were weakly connected due to the poor levels of intra-regional trade intensity and portfolio flows.

More recently, Wei et al. (2020) use the time-varying spillover model and showed evidence of volatility spillover effects among "the Belt and Road" currency market during the regional and global crises. The spillover in the RMB exchange rate was affected by internal financial reforms, as well as external economic shocks, but the spillover system for such a market was disrupted during the COVID-19 period. Based on the Correlation matrices (CM) and the information flow graph method, Mai et al. (2018) argued that, in a global context, the correlations of currency rate volatility are present in the network. They also found that the US Dollar dominates the global foreign currency markets while the Euro greatly impacts European currencies. The East Asian currencies were more strongly correlated than the European currencies due to the strong co-movement of currencies in the East Asian region.

By implementing the four major cryptocurrencies (Bitcoin, Ethereum, Ripple, and Litecoin), Kumar and Anandaramo (2019) found evidence of volatility spillover from Bitcoin to Ethereum and Litecoin, using the multivariate IGARCH-DCC and the pairwise wavelet cross-spectral analysis. In the European region, Antonakakis (2012) found evidence of return co-movements and volatility spillovers between major foreign currency rates before the introduction of the Euro. Yet, this evidence is weak in the post-euro period. Specifically, the Euro was a net transmitter of volatility, while the British pound was a net receiver of volatility in both periods. Using the GARCH and BVAR models, Orlowski (2016) showed evidence of high positive co-movements between the Euro and the non-Euro

currencies, suggesting a strong substitution between these currencies and the Euro in foreign exchange markets which significantly rose during the GFC. Kočenda and Moravcová (2019) reported significant differences in the extent of currency co-movements during various periods of financial distress. They concluded that all three currencies in the new EU foreign exchange markets brought hedging benefits during crisis periods but at different costs. During the crisis, volatility spillovers among currencies rose significantly, and the Hungarian currency showed a leading role.

Using the LASSO-VAR approach to examine the volatility connectedness among 65 major global currencies, Wen and Wang (2020) found that the US dollar and Euro exhibit volatility transmission, while other currencies, including the Japanese yen and British pound, exhibit evidence of volatility receivers. They argued that volatility connectedness responded sensitively to changes in international economic fundamentals and increased during crisis periods (oil price crashes, exchange rate regimes, and monetary policy changes). However, the directional volatility connectedness of the Renminbi significantly declined after the reforms of the Chinese exchange rate regime (which shifts from a USD-pegged exchange rate regime to a managed floating exchange rate regime).¹ Fung et al. (2022) found evidence of volatility persistence with negative leverage effects among cryptocurrencies' return behavior using the GARCH family models.

By analyzing the connectedness between six traditional currencies, and six cryptocurrencies using the quantile cross-spectral approach, Baumöhl (2019) showed evidence of significant negative dependencies between currencies and cryptocurrencies in both the short- and long-term horizons, suggesting that investors can diversify their portfolio by using a mixture of these two asset groups. By examining the effect of variation in the Chinese currency regulatory management system on the co-movement between the Chinese currency (RMB) with regional and other emerging market currencies after the three post-reform periods of RMB management (transition, basket of currencies management, and countercyclical management), McCauley and Shu (2019) revealed that the co-movement with regional and Latin American currencies peaked in the basket period while it declined after the countercyclical period. Using a diagonal BEKK model, Hsu et al. (2021) argued that cryptocurrencies acted as a safe haven during the COVID-19 pandemic against risk spillover.

The other strand of studies deals with the effect of economic policy uncertainty and/or trade uncertainty on currencies. In general, previous studies on this issue did not succeed in presenting clear-cut evidence of the dynamic connectedness between currencies and Bitcoin in the presence of economic policy uncertainty before and during the COVID-19 pandemic. Mokni et al. (2020) argued that EPU is negatively associated with the dynamic conditional correlations between Bitcoin and the US stock markets only after the crash of Bitcoin in December 2017, highlighting the usefulness of using Bitcoin as a hedging instrument to the stock portfolio. By exploring the connectedness among the nine US dollar exchange rates of globally traded currencies in association with the effect of trade policy uncertainty, Huynh, Nasir, and Nguyen (2020) found evidence of connectedness among the US dollar exchange rates when trade policy uncertainty is present. Regarding the Chinese case, Chen et al. (2020) reported that economic policy uncertainty positively affects China's exchange rate volatility on all quantiles.

Alam et al. (2019) argued that oil prices are strongly connected to the exchange rates of six major bilateral currencies against the US dollar during the GFC period and the European sovereign debt crisis. Further, Mokni and Ajmi (2020) provided evidence of the linkage between the top-five cryptocurrencies and the US dollar under different market conditions, especially before and during the COVID-19 period. During the COVID-19 crisis, the connectedness between the US dollar and cryptocurrencies was discovered at higher and lower tails of the distribution, and the predictive power of the US dollar was lost in favor of cryptocurrencies, leading cryptocurrencies to be good predictors and acting as hedging instruments against the US dollar variation. When adopting the Quantile-VAR approach, Al-Shboul et al. (2022) argued that the impact of cryptocurrency policy and price uncertainties on the dynamic connectedness among cryptocurrencies rose widely during the COVID-19 outbreak. Bouri et al. (2021) also found that the connectedness among cryptocurrencies in the lower and upper quantiles is substantially higher than those in the mean and median of the conditional distribution, confirming that connectedness increases with an increase in shock size either for both positive or negative shocks. Another study by Kumar et al. (2022) reported evidence of a presence of structural change in the connectedness evolving in 2020 as a result of the COVID-19-pandemic.

Overall, the above-referenced existing studies have not fully emphasized the connectedness among currencies, especially major global currencies. Although the existing studies have examined the effect of policy factors (e.g., exchange rate regimes, central bank interventions under inflation, and capital controls), and economic factors (e.g., financial crises, international trade, oil exports, investor behavior, and trade deficits) on volatility transmission across currencies, they failed to provide a comparative analysis between connectedness among the two types of currencies as an example. On the other hand, previous studies in this context have overlooked the effect of economic uncertainty. More precisely, they failed to report clear-cut evidence of the direction of the relationship between economic policy uncertainty and exchange rates. Few studies concluded that economic policy uncertainty has a negative effect on currency markets, while others argued that economic policy uncertainty positively impacts currency markets. Second, there is a general paucity in the literature concerning the impact of economic uncertainty on the connectedness among currencies during normal and distressed market conditions. Third, although the impact of economic policy uncertainty on currency markets is well documented in the literature, existing studies have noticeably overlooked the examination of the effects of economic uncertainty on the connectedness among major traditional currencies with leading cryptocurrencies during the COVID-19 period where significant currencies fluctuations and high recessions have emerged. Lastly, most of them focused on a small number of

¹ Another strand of literature focused on the volatility connectedness between traditional currencies and cryptocurrencies. For example, Andrada-Félix et al. (2020) found evidence of volatility connectedness across cryptocurrencies and traditional currencies. The volatility connectedness was found to be time-varying, especially during the periods of economic and financial instability, due to idiosyncratic shocks in the markets.

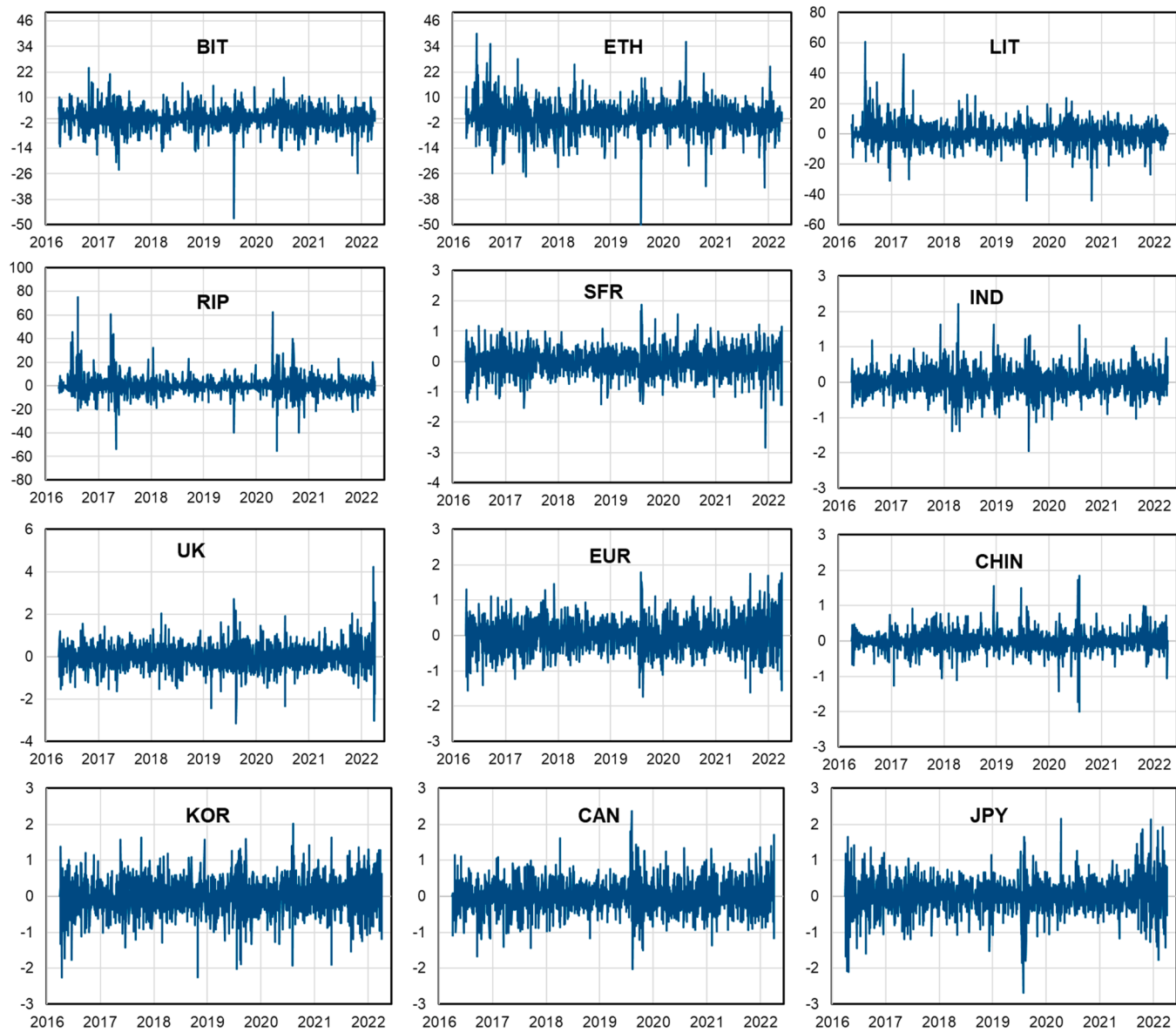


Fig. 1. Return series of conventional and digital currencies.

Table 1

Descriptive Statistics and preliminary analysis before COVID-19.

	BIT	ETH	LIT	RIP	SFR	IND	UK	EUR	CHIN	KOR	CAN	JPY
<i>Descriptive statistics and preliminary tests</i>												
Mean	0.254	0.380	0.290	0.417	-0.010	0.010	-0.005	-0.008	0.000	-0.001	0.003	-0.013
Variance	24.559	46.774	56.749	71.678	0.142	0.120	0.249	0.152	0.066	0.218	0.150	0.202
Skewness	-0.035	0.485 **	1.684 ***	2.074 ***	-0.346 ***	0.579 ***	-0.173 **	-0.198 **	0.346 ***	-0.441 ***	-0.103	-0.646 ***
Ex.Kurtosis	3.035 ***	4.072 ***	11.382 ***	16.403 ***	1.317 ***	3.867 ***	1.451 ***	1.022 ***	4.871 ***	2.186 ***	1.839 ***	3.661 ***
JB	187.700 ***	98.840 ***	61.694 ***	1374.614 ***	2.141	77.005 ***	11.593 ***	9.034 **	166.883 ***	7.194 **	94.223 ***	516.348 ***
ADF	-24.911***	-24.581***	-24.148***	-23.586***	-21.414***	-23.745***	-22.986***	-21.923***	-25.717***	-25.002***	-23.541***	-22.604***
PP	-24.927***	-24.634***	-24.161***	-23.637***	-21.491***	-23.736***	-22.971***	-21.907***	-25.649***	-25.015***	-23.543***	-22.652***
ERS	-4.907 **	-9.783 ***	-6.330 ***	-9.496 **	-13.177 ***	-11.611 ***	-12.317 ***	-12.922 ***	-11.421 **	-13.330 ***	-12.402 ***	-12.460 ***
Q(10)	4.737	10.493 *	9.805 *	41.847 **	8.448	10.418 *	6.520	5.002	12.278 *	2.597	3.972	3.820
Q2(10)	35.568 **	30.114 ***	68.180 ***	82.492 **	13.362 **	28.954 ***	4.424	19.715 ***	11.143 **	10.873 **	8.156	114.930 ***
<i>Correlation matrix</i>												
BIT	1.000											
ETH	0.619	1.000										
LIT	0.589	0.596	1.000									
RIP	0.431	0.424	0.452	1.000								
SFR	-0.021	-0.066	0.011	0.028	1.000							
IND	-0.045	-0.060	-0.041	-0.086	0.077	1.000						
UK	-0.026	-0.020	-0.005	0.000	0.381	0.166	1.000					
EUR	-0.008	-0.034	0.037	0.020	0.744	0.182	0.502	1.000				
CHIN	0.049	0.015	0.005	-0.009	0.218	0.314	0.275	0.349	1.000			
KOR	-0.031	-0.022	-0.011	-0.068	0.194	0.414	0.202	0.326	0.466	1.000		
CAN	-0.055	-0.093	-0.083	-0.075	0.259	0.193	0.298	0.362	0.302	0.321	1.000	
JPY	0.004	-0.012	0.036	0.017	0.573	0.009	0.210	0.400	0.114	0.072	0.100	1.000

Notes: This table reports the descriptive statistics of currencies (cryptocurrencies and traditional currencies) before the COVID-19 pandemic period. JB is the Jarque-Bera normality test statistics. To test for stationary, the augmented Dickey-Fuller (ADF), the Phillip-Perron (PP) and the Elliot-Rothenberg-Stock (ERS) unit root tests are used. Q(20) and Q2(20) are the Ljung-Box tests for 20th-order serial correlations for returns and squared returns, respectively. LM(20) is the LM heteroscedasticity test at order 20. (***), (**), and (*) indicate the statistical significance, respectively, at the 1%, 5%, and 10% levels.

currencies (mostly hard currencies) but ignored the impact of economic policy uncertainty on the connectedness among a mix of hard and emerging countries' currencies.

3. Data and methodology

3.1. Data

To conduct our study, we collect a database on the daily exchange rates of twelve currencies, consisting of four cryptocurrencies and eight traditional currencies. The cryptocurrencies group includes Bitcoins (BIT), Ethereum (ETH), Litecoin (LIT), and Ripple (RIP), while the traditional currencies are: the Swiss Franc (SFR), Indian Rupee (IND), the Euro (EUR), Great Britain Pound (UK), Chinese Yuan (CHIN), Korean Won (KOR), Canadian Dollar (CAD), and the Japanese Yen (JPY). All the data covers the period from January 2, 2017, to October 6, 2022. The choice of the sample period accounts for two main issues: *i*) the significant drop in the value of cryptocurrencies after the beginning of 2018 and currency markets indices in general, as a result of the uncertainty generated by the Japanese cryptocurrency exchange Coincheck getting hacked, and was robbed to the tune of half a billion dollars; and *ii*) the uncertainty created by the ongoing COVID-19 crisis period. The traditional and digital currencies under study are taken based on their equivalency to the US dollar. Traditional currencies are collected from Refinitiv (DataStream database), while cryptocurrencies are obtained from Coinmarket.com. To conduct our analysis, we use the percentage log-returns.

To examine the possible effect of the COVID-19 pandemic crisis on the connectedness between currencies, our sample period is divided into two sub-periods based on the declaration of the World Health Organization (WHO) that COVID-19 is a pandemic on March 11th, 2020. To investigate the effect of economic policy uncertainty (EPU) on the connectedness among currencies, we use three global risk indices (EPU, the fear index (VIX), which is the CBOE volatility index, and (OVX) representing the crude oil market volatility). The proxy for EPU is obtained from The Economic Policy Uncertainty index website: www.policyuncertainty.com. This index is developed by [Baker et al. \(2016\)](#) and reflects the growing uncertainty-generated policies that impact economic policy and financial decisions. The data of the other proxies for market uncertainties, such as VIX and OVX, were downloaded from the investing.com website. These indices have been used by several studies (among others, [Manela and Moreira, 2017](#); [Fang et al., 2020](#); [Gozgor et al., 2019](#)).

[Fig. 1](#) shows the exchange rates and cryptocurrencies' returns series during the sample period. We observe increased volatility and volatility clustering, especially during the COVID-19 pandemic period for almost all series. However, returns series of Chinese Yuan, Bitcoin, Ethereum and Ripple show less volatility during the pandemic. [Tables 1 and 2](#) provide summary statistics for the returns series before and during the COVID-19 periods, respectively. Generally, cryptocurrencies show positive average returns before the crisis, whereas traditional currencies show mixed signs of average returns. We see that Indian Rupee, the Canadian dollar and Chinese Yuan show positive average return while the Swiss Franc, British Pound, Euro, Korean Won, and Japanese Yen, show average negative returns. However, the results differ during the pandemic crisis in which traditional currencies and cryptocurrencies exhibit positive average returns. During the crisis, there was a noticeable change in the level of standard deviation for all currencies' series, suggesting that both types of currencies are characterized by different levels of risk. More specifically, cryptocurrencies show higher standard deviations than traditional currencies during the crisis. In general, cryptocurrencies experience a significant drop in value after the beginning of 2018, leading to lower returns.

Considering the COVID-19 period, we notice that cryptocurrencies have the highest positive returns (Ethereum has the highest return and Litecoin shows the lowest positive return). However, for traditional currencies, the Swiss Franc, Canadian dollar, and Chinese Yuan have the lowest returns, while the British Pound, Korean Won, and Japanese Yen experienced the highest positive returns. Then, testing for normality, skewness, and excess kurtosis values indicate that the return series are asymmetric and fat-tailed. The Jarque-Berra test statistics are all significant at the 1% level for all series, confirming the non-normality issue.

Finally, the results from the augmented Dickey-Fuller (ADF) ([Dickey and Fuller, 1979](#)), the Phillip-Perron (PP) ([Phillips and Perron, 1988](#)) and the Elliot-Rothenberg-Stock (ERS) ([Elliot et al., 1996](#)) unit root tests indicate that all return series are stationary. Moreover, the Ljung-Box Q(10) and Q2(10) tests of the autocorrelation of returns and squared returns series indicate that the null hypothesis of no autocorrelation is rejected in almost of cases at the 5% significance level. [Tables 1 and 2](#) also provide the correlation matrix between all currency return series. The correlations between the two types of currencies differ across both sub-periods; correlations between cryptocurrencies and traditional currencies are generally negative during the COVID-19 period. However, cryptocurrencies, as a separate group, are positively correlated in both sub-periods. Traditional currencies, as a separate group, are also significantly and positively correlated with each other during the COVID-19 period. The negative correlations between cryptocurrencies and traditional currencies during COVID-19 indicate the usefulness of both types of currencies to hedge each other during stress periods.

3.2. Methodology

This section shows the econometric methodology used in the empirical analyses of the total, net and directional connectedness between cryptocurrencies and traditional currencies. Initially, we outline the methodology proposed by [Diebold and Yilmaz \(2009, 2012, 2014\)](#) and then present the dynamic connectedness procedure based on the TVP-VAR methods proposed by [Antonakakis and Gabauer \(2017\)](#). This approach is proven to gain a lot of attention and popularity, as evidenced by many studies, including those by [Korobilis and Yilmaz \(2018\)](#), [Gabauer and Gupta \(2018\)](#), [Liu and Gong \(2020\)](#), [Youssef et al. \(2021\)](#), and [Mokni et al. \(2021\)](#), among others.

Table 2

Descriptive Statistics and preliminary analysis during COVID-19.

	BIT	ETH	LIT	RIP	SFR	IND	UK	EUR	CHIN	KOR	CAN	JPY
<i>Descriptive statistics and preliminary tests</i>												
Mean	0.138	0.290	0.017	0.130	0.008	0.016	0.021	0.021	0.003	0.025	0.000	0.048
Variance	23.019	40.981	40.066	58.436	0.215	0.116	0.399	0.228	0.086	0.264	0.229	0.231
Skewness	-1.659 ***	-1.145 ***	-1.278 ***	0.245 ***	-0.220 **	0.123	0.385 ***	0.326 ***	0.000	-0.102	0.328 ***	0.401 ***
Ex.Kurtosis	15.367 ***	12.470 ***	8.087 ***	14.650 ***	2.754 ***	3.171 ***	5.815 ***	1.382 ***	9.888 ***	1.293 ***	1.639 ***	2.768 ***
JB	6909.852 ***	4494.322 ***	2011.299 ***	6007.306 ***	217.516 ***	282.884 ***	961.928 ***	65.238 ***	2733.803 ***	47.885 ***	87.146 ***	232.227 ***
ADF	-18.933***	-18.685***	-17.889***	-15.678***	-14.602	-17.884***	-15.835***	-13.932***	-21.789***	-16.667***	-15.856***	-14.345***
PP	-18.949***	-18.824***	-17.886***	-15.6225***	-14.098***	-17.917***	-15.891***	-13.992***	-22.829***	-16.687***	-15.893***	-14.210***
ERS	-0.504	-0.524	-0.730	-0.845	-1.361	-2.064 **	-1.040	-1.284	-1.189	-2.065 *	-2.437 *	-3.015 ***
Q(10)	13.143 **	16.265 ***	13.701 **	4.576	6.127	5.872	10.904 **	7.372	25.492 ***	18.604 ***	9.300 *	2.920
Q2(10)	3.770	12.319 **	18.941 ***	21.068 ***	14.500 ***	38.153 ***	128.080 ***	60.823 ***	147.701 ***	66.687 ***	50.816 ***	93.721 ***
<i>Correlation matrix</i>												
BIT	1.000											
ETH	0.827	1.000										
LIT	0.829	0.831	1.000									
RIP	0.560	0.592	0.654	1.000								
SFR	-0.061	-0.054	-0.039	-0.003	1.000							
IND	-0.049	-0.050	-0.043	0.014	0.319	1.000						
UK	-0.095	-0.085	-0.062	-0.029	0.530	0.409	1.000					
EUR	-0.099	-0.084	-0.078	-0.021	0.740	0.403	0.630	1.000				
CHIN	-0.104	-0.065	-0.071	-0.001	0.357	0.308	0.417	0.352	1.000			
KOR	-0.042	-0.018	0.003	0.022	0.361	0.375	0.460	0.468	0.339	1.000		
CAN	-0.089	-0.063	-0.056	-0.024	0.433	0.446	0.626	0.537	0.344	0.487	1.000	
JPY	-0.001	0.017	0.019	0.025	0.530	0.132	0.351	0.405	0.202	0.189	0.177	1.000

Notes: This table reports the descriptive statistics of currencies (cryptocurrencies and traditional currencies) before the COVID-19 pandemic period. JB is the Jarque-Bera normality test statistics. To test for stationary, the augmented Dickey-Fuller (ADF), the Phillip-Perron (PP) and the the Elliot-Rothenberg-Stock (ERS) unit root tests are used. Q(20) and Q2(20) are the Ljung-Box tests for 20th-order serial correlations for returns and squared returns, respectively. LM(20) is the LM heteroscedasticity test at order 20. (***), (**), and (*) indicate the statistical significance, respectively, at the 1%, 5%, and 10% levels.

3.2.1. TVP-VAR approach

Researchers have developed methods to capture shock transmission mechanisms between macroeconomic variables. Diebold and Yilmaz (2009, 2012, 2014) (DY) established a rolling-window VAR-based empirical approach providing various connectedness measures built from pieces of variance decompositions. Their approach assesses forecast error variation in the variable i attributed to innovations in other variables in the model. They considered that the time series follows the reduced-form autoregressive (VAR) model with a fixed parameter and variance-covariance matrix.

Alternatively, to enhance the accuracy of the dynamic connectedness measures of DY, Antonakakis and Gabauer (2017) used a time-varying parameter vector autoregressive model (TVP-VAR) with a time-varying covariance structure as proposed in Primiceri (2005). They adopted this model in order to challenge the constant-parameter rolling-window VAR approach. The drifting coefficients and stochastic volatility are employed to obtain i) the possible nonlinearities or time variation in the lag structure of the model, ii) the possible heteroscedasticity of the shocks, and iii) the nonlinearities in the simultaneous relations among the variables of the model (Primiceri, 2005). The TVP-VAR approach improves the Diebold and Yilmaz methodology in many ways. It allows “to capture possible changes in the underlying structure of the data in a more flexible and robust manner” as in Antonakakis and Gabauer (2017) and Antonakakis et al. (2020). This indicates that the TVP-VAR estimations are superior to those generated by rolling-windows. First, since the rolling window size is not set arbitrarily, there is no loss of information in calculating the dynamic measures of connectedness. Second, as the rolling window size is based on a multivariate Kalman filter, it could be less sensitive to the presence of outliers and thus adjusts immediately to events (e.g., Antonakakis et al., 2019; Gabauer and Gupta, 2018).

The N -variable TVP-VAR(p) model can thus be written as follows,

$$Y_t = \beta_t Z_{t-1} + \varepsilon_t, \varepsilon_t | \Omega_{t-1} \sim N(0, \Sigma_t) \quad (1)$$

$$\beta_t = \beta_{t-1} + \vartheta_t, \vartheta_t | \Omega_{t-1} \sim N(0, R_t) \quad (2)$$

where $Y_t = (Y_{1t}, Y_{2t}, \dots, Y_{Nt})'$ is a vector with N variables and Z_{t-1} is an $N_p \times 1$ conditional vector formed of the past p lags, p being the optimal lag length with $Z_{t-1} = (Y_{t-1}, Y_{t-2}, \dots, Y_{t-p})'$, and Ω_{t-1} represents all information available through time $t - 1$, whereas β_t is an $N \times N_p$ dimensional time-varying coefficient matrix, which follows a random walk model and can be represented as $\beta_t = (\beta_{1t}, \beta_{2t}, \dots, \beta_{pt})$, β_{it} being an $N \times N$ matrix of time-varying coefficients.

The error-disturbance vectors ε_t and ϑ_t are $N \times 1$ and $N \times N_p$ dimensional with $N \times N$ and $N_p \times N_p$ time-varying variance-covariance matrices, Σ_t and R_t respectively.

The Kalman filter algorithm is employed with forgetting factors chosen based on a Bayesian model selection, as introduced by Koop and Korobilis (2014) and demonstrated in Antonakakis et al. (2019).

To calculate the generalized forecast error variance decompositions (GFEVD), the model in Eq. (1) is transformed to its vector moving average (VMA) representation based on the Wold representation theorem. The representation of the system is:

$$Y_t = \sum_{j=0}^{\infty} \Theta_{jt} \varepsilon_{t-j}, \quad (3)$$

where Θ_{jt} is an $N \times N$ dimensional matrix.

3.2.2. Dynamic connectedness measures

To measure the dynamic connectedness between different variables, the time-varying parameters and variance-covariance matrices of the TVP-VAR model are used in Diebold and Yilmaz's measure of connectedness. The elements of the dynamic H -step generalized variance decomposition matrix $D_t^{gH} = [d_{ij,t}^{gH}]$ are defined as:

$$d_{ij,t}^{gH} = \frac{\sigma_{jj,t}^{-1} \sum_{h=0}^{H-1} (e'_i \Theta_{h,t} \Sigma_t e_j)^2}{\sum_{h=0}^{H-1} (e'_i \Theta_{h,t} \Sigma_t \Theta'_{h,t} e_j)},$$

where $\sigma_{jj,t}$ is the j^{th} diagonal element of Σ_t . The normalized terms $\tilde{d}_{ij,t}^{gH} = \frac{d_{ij,t}^{gH}}{\sum_{j=1}^N d_{ij,t}^{gH}}$ are used to determine the dynamic total directional connectedness, net total directional connectedness, and total connectedness as follows.

The *total connectedness index* (TCI), which measures interconnectedness among the time-series, is calculated as:

$$C_t^{gH} = \frac{\sum_{i,j=1, i \neq j}^N \tilde{d}_{ij,t}^{gH}}{\sum_{j=1}^N \tilde{d}_{ij,t}^{gH}} \times 100 \quad (4)$$

The *directional spillover* received by variable i from all other variables j , is measured as:

Table 3

Total average dynamic connectedness.

	BIT	ETH	LIT	RIP	SFR	IND	UK	EUR	CHIN	KOR	CAN	JPY	FROM
<i>Before COVID-19</i>													
BIT	35.558	21.942	20.438	15.908	0.625	0.98	0.627	1.031	0.404	0.689	0.5	1.298	64.442
ETH	21.247	33.56	21.648	17.225	0.614	1.186	0.663	0.967	0.508	0.827	0.715	0.841	66.44
LIT	20.219	23.841	33.226	15.816	0.865	0.991	0.9	1.372	0.375	0.856	0.461	1.076	66.774
RIP	17.372	20.93	17.439	37.2	0.761	1.909	0.735	0.793	0.46	0.68	0.664	1.057	62.8
SFR	1.252	0.713	0.703	0.546	49.382	1.387	5.705	23.645	1.842	1.844	3.122	9.859	50.618
IND	0.666	0.775	1.033	1.598	1.282	63.341	2.823	2.783	8.962	12.794	3.032	0.912	36.659
UK	1.416	1.014	1.257	0.681	6.452	2.826	57.758	13.813	3.573	3.041	6.657	1.512	42.242
EUR	0.734	1.387	1.219	0.666	21.368	1.968	10.847	43.422	5.264	4.783	5.72	2.62	56.578
CHIN	1.555	0.704	0.518	1.421	2.457	7.068	3.445	5.748	54.583	14.816	6.826	0.861	45.417
KOR	0.836	0.866	0.929	0.749	1.716	9.668	1.839	5.057	14.588	53.977	8.024	1.752	46.023
CAN	0.717	0.788	0.434	0.655	2.936	2.156	5.712	9.47	8.8	8.503	58.253	1.575	41.747
JPY	2.168	0.691	0.739	0.852	15.678	1.502	2.716	4.845	1.703	1.119	0.761	67.227	32.773
Cont TO others	68.183	73.65	66.355	56.117	54.755	31.642	36.011	69.524	46.478	49.951	36.481	23.364	612.51
Cont incl own	103.741	107.211	99.581	93.317	104.137	94.983	93.769	112.947	101.061	103.929	94.734	90.591	TCI
Net spillovers	3.741	7.211	-0.419	-6.683	4.137	-5.017	-6.231	12.947	1.061	3.929	-5.266	-9.409	51.043
<i>During COVID-19</i>													
BIT	35.55	22.62	23.96	11.35	0.52	0.59	1.13	1	0.8	0.66	1.5	0.32	64.45
ETH	22.54	35.73	23.71	14.04	0.36	0.48	0.86	0.63	0.2	0.31	0.9	0.25	64.27
LIT	22.98	22.88	34.14	15.04	0.55	0.39	0.83	0.84	0.45	0.42	1.13	0.33	65.86
RIP	14.25	17.58	19.65	45.57	0.24	0.34	0.41	0.31	0.23	0.4	0.6	0.43	54.43
SFR	0.7	0.72	0.91	0.42	36.07	3.85	9.2	21.22	4.75	5.16	6.37	10.63	63.93
IND	1.06	1.46	1.44	1.38	5.03	49.9	8.43	8.34	5.33	6.6	9.69	1.34	50.1
UK	1.74	1.86	1.63	1.07	9.19	6.18	37.67	11	5.59	7.01	13.37	3.7	62.33
EUR	1.22	1.34	1.47	0.59	19.93	5.61	10.27	34.3	4.23	6.27	8.37	6.41	65.7
CHIN	1.21	1.09	1.19	0.71	6.8	4.97	7.79	6.46	55.63	5.15	5.86	3.15	44.37
KOR	1.59	1.92	1.66	1.51	5.64	5.86	9.18	7.67	4.88	48.91	9.55	1.63	51.09
CAN	2.27	3.02	2.14	1.36	6.21	7.87	14.12	8.76	5.01	8.14	40.25	0.84	59.75
JPY	1.16	0.84	1.05	0.96	15.96	2.19	5.81	10.44	3.62	2.51	1.56	53.9	46.1
Cont TO others	70.72	75.33	78.8	48.44	70.42	38.33	68.01	76.69	35.1	42.62	58.9	29.03	692.38
Cont incl own	106.27	111.06	112.94	94.01	106.49	88.23	105.68	110.99	90.73	91.53	99.15	82.93	TCI
Net spillovers	6.27	11.06	12.94	-5.99	6.49	-11.77	5.68	10.99	-9.27	-8.47	-0.85	-17.07	57.7

Notes: This table reports the results of the average connectedness measures based on the [Diebold and Yilmaz \(2012, 2014\)](#) estimated from a TVP-VAR model for the full sample before and during the COVID-19 period. We provide the average of total connectedness, the directional spillover received (denoted by “From”), and transmitted (denoted by “Contribution to others”) by each variable. Then, the net directional spillover (denoted by “Net Spillovers”) is obtained as the difference between directional ‘To’ spillovers and directional ‘From’ spillovers.

$$C_{i \leftarrow j}^{gH} = \frac{\sum_{j=1, i \neq j}^N \tilde{d}_{ij,t}^{gH}}{\sum_{i=1}^N \tilde{d}_{ij,t}^{gH}} \times 100 \quad (5)$$

Likewise, the spillovers received by variable j from all other variables i , is measured as:

$$C_{i \rightarrow j}^{gH} = \frac{\sum_{j=1, i \neq j}^N \tilde{d}_{ij,t}^{gH}}{\sum_{j=1}^N \tilde{d}_{ij,t}^{gH}} \times 100 \quad (6)$$

In order to obtain the *net pairwise directional connectedness*, we subtract the total directional connectedness to others from the total directional connectedness from others, which can be interpreted as the influencing variable i has on the analyzed network. That is,

$$C_{ij,t}^{gH} = C_{j \leftarrow i,t}^{gH} - C_{i \leftarrow j,t}^{gH}. \quad (7)$$

At last, the *net pairwise directional connectedness* is defined as:

$NPDC_{ij}^{gH} = (\tilde{d}_{ji,t}^{gH} - \tilde{d}_{ij,t}^{gH}) \times 100$. A result greater than zero indicates that variable i dominates variable j ; otherwise, the latter is said to dominate.

4. Empirical analysis

In this study, we perform two empirical analyses. First, the dynamic connectedness among currencies (both types of currencies: traditional and cryptocurrencies) is quantified using the procedure introduced by Diebold and Yilmaz (2009, 2012, 2014) based on the time-varying parameter vector autoregressive (TVP-VAR) methodology. Second, the effect of economic policy uncertainty (EPU) on this connectedness is estimated based on a quantile regression model.

4.1. Dynamic connectedness results

Based on the TVP-VAR approach, the results of total connectedness among all currencies before and during the health crisis are reported in Table 3. The table shows that the total connectedness index (TCI) exhibits different behaviors before and during the pandemic. Indeed, the average TCI increased from 51.043 before the pandemic to 57.7 during the crisis. Fig. 2 depicts the dynamic TCI and provides further results supporting those reported in Table 3. As shown in Fig. 2, total connectedness increased at the beginning of the COVID-19 crisis. It fell somehow during 2021 due to the drop in death cases based on the discovery of the vaccination, but then it started to increase during the 2022 period, marking the impact of the Russian-Ukraine war.

Further, Table 3 shows the contribution of each currency to others before the COVID-19 pandemic and emphasizes that cryptocurrencies are the most transmitters of shocks compared to traditional currencies. Ethereum is the highest transmitter, while Ripple is the lowest one. The highest traditional currencies transmitters are the Swiss Franc and the Euro, while the lowest transmitters are the Indian Rupee and the Japanese Yen. Although the level of volatility transmission rose significantly across all types of currencies during the pandemic crisis, the majority of traditional currencies are the highest transmitters (the Euro (76.69), the British Pound (UK) (68.01), the Swiss Franc (SFR) (70.42), the Canadian Dollar (CAD) (58.9), followed by the cryptocurrencies (Bitcoin (70.72), Ethereum (75.33) and Litecoin (78.8)). These results confirm that the dominance of currencies in the shock transmission changes from one currency to another before and during COVID-19. Traditional currencies (EUR, UK, SFR, and CAN) as well as cryptocurrencies (BIT,

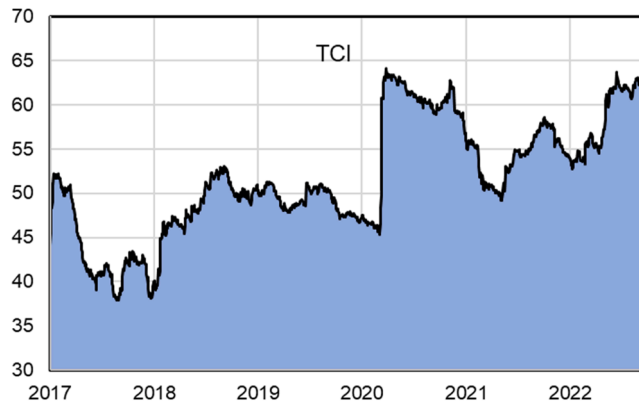


Fig. 2. Total connectedness between currencies over the full period.

ETH, and LIT) are the dominant transmitters of shocks during the crisis, with the Euro, British Pound, and Swiss Franc being the most interconnected with other currencies.²

Regarding the amount of shocks received by each currency from the others, Table 3 shows that the most receiver of shocks from others is almost all cryptocurrencies, as well as a few traditional currencies, namely: Euro (56.57) and Swiss Franc (50.61) before the COVID-19 period, and 62.55 and 63.93, for the during COVID-19 period. During the health crisis, although the same cryptocurrencies maintained the same position as the highest receivers of shocks from the system, traditional currencies became the highest receivers as well. Therefore, during the crisis, almost all currencies, either cryptocurrencies or traditional currencies, became more receivers of shocks. Our results are inconsistent with those obtained by Baumöhl (2019) while falling in-line with those of Chemkha et al. (2021). For example, Baumöhl (2019) analyzed the relationship between traditional currencies and cryptocurrencies and found a negative relationship between the two types of currencies in the short-and long-term. While, Chemkha et al. (2021) provided evidence that connectivity between cryptocurrencies has greatly strengthened since 2017 yet, “the dependence is positive and higher for the pairs of the same market than those across markets, and the cryptocurrency market and the fiat currency market are weakly connected,” providing investors with more benefits from holding these assets together, for portfolio diversification purposes.

Table 3 also includes the results of the net connectedness. The currency is a net transmitter (receiver) of shocks if this index is positive (negative). The net connectedness index varies over currencies before and during the COVID-19 crisis. Before the crisis, four traditional currencies (SFR, EUR, CHIN, KOR) and two cryptocurrencies (ETH and BIT) were found to be net transmitters. However, the others are found to be net receivers. The highest net transmitters are the Euro and Ethereum, whereas the most net receivers are the Japanese Yen, the British Pound, the Canadian dollar, the Indian Rupee, and Ripple as well. During the COVID-19 crisis, the highest cryptocurrency net transmitters rose to three (BIT, ETH, and LIT). Litecoin has turned from the lowest net receivers to one of the highest net transmitters compared to the period before the crisis. Such a result can be interpreted by the important position taken by Litecoin as a safe haven asset that dominates this of Bitcoin. Our results confirm the previous findings of Yi et al. (2018), Bouri et al. (2019), Ji et al. (2020), Moratis (2021), Al-Shboul et al. (2022) and Raza et al. (2021).

In addition, the results show that the level of net transmission increased significantly during the crisis since five traditional currencies (SFR, EUR, UK, CHIN, and CAD) became net transmitters to the system. In particular, the Swiss Franc, the Euro, and the British Pound are the highest net transmitters. The highest net receivers during the pandemic crisis were the Ripple, the Japanese Yen, Indian Rupee, and the Korean Won. Overall, we find an increase in the level of net transmitting and net receiving of shocks during the COVID-19 period. Our results are consistent with those of Shahzad et al. (2021), who considered four cryptocurrencies (Bitcoin, Ethereum, Ripple, and Litecoin) and four traditional currencies, namely the Euro, Japanese yen (JPY), the British pound (GBP), and Chinese yuan (CNY). During the negative explosiveness and non-bubble periods, they found that JPY is the most consistent hedger for the considered cryptocurrencies, followed by GBP and the Euro. All other currencies, except the Euro, have safe-haven properties for Bitcoin and Litecoin. Our results also fall in-line with Baumöhl (2019), who argued that Euro, Japanese Yen, and Chinese yuan share a safe-haven potential during turmoil and extreme market periods.³

We go further to obtain the total and the net spillover among currencies as separate groups. Table 4 presents the results of the dynamic connectedness among cryptocurrencies. On average, the total connectedness among cryptocurrencies during the COVID-19 crisis is higher than before the crisis period. The highest contributors to other cryptocurrencies are Ethereum and Litecoin during and before the crisis periods, with a noticeable increase in contribution levels during the pandemic period. The net spillover among cryptocurrencies indicates that Ethereum is the highest net transmitter before the crisis, while Ripple is found to be the highest net receiver. Overall, during the crisis period, cryptocurrencies show evidence of more net transmitters and net receivers than before COVID-19.

Table 5 presents the results of spillovers among traditional currencies. Interestingly, such results are different from the results of cryptocurrencies. The total spillover among traditional currencies, on average, during the COVID-19 crisis is higher compared to the period before crisis. The highest contributors to other traditional currencies are the Swiss Franc, Euro, Chinese Yuan, Korean Won, and Canadian Dollar in both sub-periods. Yet, during the crisis, there is a noticeable increase in the level of contribution of the traditional currencies compared to the period before the pandemic. We observe that only four traditional currencies are the most receivers of information. Still, the number of traditional currencies receivers is significantly increasing during COVID-19, as almost all traditional currencies exhibit a higher level of information reception.

Before the crisis, two traditional currencies were net transmitters (the Swiss Franc and the Euro), while the others were net receivers. The highest net transmitter is the Euro, whereas the Indian Rupee is the highest net receiver. However, the results of net spillover are somewhat different during the COVID-19 period. The net transmitters are the Swiss Franc, the British Pound, the Euro, and the Canadian Dollar, with the Euro being the highest transmitter. The British Pound and Canadian Dollar change their position from being net receivers before the crisis to net transmitters during the crisis. The other currencies maintain their positions as net

² When analyzing the spillover effects based on the TVP- VAR approach, we made the results using several window sizes and forecasting periods. However, the results remain similar, indicating the robustness of our results. Also, we obtained all the pair-wise connectedness graphs for the full sample, before COVID-19 and during COVID-19 periods. To save space, we have not reported them, yet, they are available upon request.

³ Based on the request of a referee, we also obtain the extreme spillover based on the QVAR as suggested by Bouri et al. (2021). The results are reported in Appendix B. As can be seen, the results are similar for the before COVID-19 and during COVID-19 periods. The total connectedness at the extreme quantiles are in the range of 86%, while at the median level, they are 52%, 47%, and 56% for the full sample, before COVID-19, and COVID-19 periods, respectively.

Table 4

Total average dynamic connectedness for cryptocurrencies.

	BIT	ETH	LIT	RIP	FROM
<i>Full sample</i>					
BIT	45.38	21.34	20.47	12.8	54.62
ETH	20.51	43.27	21.71	14.51	56.73
LIT	20.06	20.94	43.44	15.56	56.56
RIP	14.17	17.39	18.3	50.15	49.85
Cont TO others	54.74	59.67	60.48	42.87	217.76
Contr incl own	100.12	102.95	103.92	93.02	TCI
Net spillovers	0.12	2.95	3.92	-6.98	54.44
<i>Before COVID-19</i>					
BIT	52.39	19.28	17.28	11.05	47.61
ETH	17.86	50.24	18.79	13.11	49.76
LIT	16.82	19.03	51.07	13.08	48.93
RIP	11.74	15.34	15.83	57.09	42.91
Cont TO others	46.41	53.65	51.91	37.24	189.21
Contr incl own	98.8	103.88	102.98	94.34	TCI
Net spillovers	-1.2	3.88	2.98	-5.66	47.30
<i>During COVID-19</i>					
BIT	38.19	24.27	25.77	11.77	61.81
ETH	23.51	37.26	24.71	14.52	62.74
LIT	24.31	24.11	36	15.59	64
RIP	14.41	18.08	20.14	47.37	52.63
Cont TO others	62.22	66.46	70.63	41.88	241.19
Contr incl own	100.41	103.72	106.63	89.25	TCI
Net spillovers	0.41	3.72	6.63	-10.75	60.30

Notes: This table reports the results of the average connectedness measures of cryptocurrencies based on the Diebold and Yilmaz (2012, 2014) estimated from a TVP-VAR model for the full sample before and during the COVID-19 period. We provide the average of total connectedness, the directional spillover received (denoted by “From”), and transmitted (denoted by “Contribution to others”) by each variable. Then, the net directional spillover (denoted by “Net Spillovers”) is obtained as the difference between directional ‘To’ spillovers and directional ‘From’ spillovers.

receivers of shocks during the pandemic period. The highest net receivers are the Indian Rupee and the Japanese Yen. Overall, traditional currencies display a higher level of net spillover during the COVID-19 period.

The results from Tables 3, 4, and 5 are confirmed by the graphs reported in Figs. 3–5. Fig. 3 provides the behavior of the net dynamic spillover in the full sample periods for both groups of currencies. Moreover, Fig. 4 provides the dynamic net connectedness indices before and over the COVID-19 pandemic period. A close look at this figure shows that the net spillover among currencies is higher during the COVID-19 period than before the crisis period for all currencies. Due to the COVID-19 crisis in March 2020, the spillover level got higher. For traditional currencies, the net spillover shows higher levels of net transmission and receiving of shocks, implying that during the COVID-19 period, the net connectedness among traditional currencies increased. However, the net spillover among cryptocurrencies at the beginning of the crisis showed a lower level of net transmission and net receiving than before the crisis, indicating a significant increase at the end of 2020.

For the pre-COVID-19 period, Fig. 4 shows a higher fluctuation in the net spillover of all cryptocurrencies and traditional currencies before the crisis compared to their situation during the crisis. However, the net spillover among cryptocurrencies shows higher levels of net transmission and net reception of shocks during the crisis than before the crisis. The net spillover of cryptocurrencies has a wider range than traditional currencies during the COVID-19 crisis. As indicated by Fig. 5, during the crisis period, we notice a wider range of net spillover of all currencies in the early beginning of the pandemic, and such range started to relax at the end of the third quarter of 2020, with the beginning of the vaccination programs. However, after the end of 2020, there was an increase in the net spillover among all currencies. In light of the traditional currencies, we see an increase in the net spillover after the third quarter of 2020. Although the net spillover of the traditional currencies became wider, the level of net spillover of cryptocurrencies became higher than the net spillover of the traditional currencies during the crisis period.

4.2. EPU and dynamic connectedness

After obtaining the spillover measures, our empirical analysis provides further evidence of the relationship between economic policy uncertainty and connectedness among currencies. We use mainly Economic Policy Uncertainty (EPU). The EPU was developed by Baker et al., (2016, 2012), respectively, due to the growing uncertainty-generated policies that impact economic policy and financial decisions. These relate to oil price fluctuations, regulatory conflicts, disputes over income distribution inequality, etc., that are happening everywhere. To further support our analysis, we also adopt two other uncertainty factors, namely, the CBOE fear index (VIX), and the crude oil volatility (OVX).⁴

To examine this relationship, we run a quantile regression model to obtain the effects of these factors on the dynamic (total and net)

⁴ We thank a referee for the suggestion of including VIX and OVX in the regression analysis.

Table 5

Total dynamic connectedness for traditional currencies.

	SFR	IND	UK	EUR	CHIN	KOR	CAN	JPY	FROM
<i>Full sample</i>									
SFR	41.38	2.06	8.17	23.09	4.56	3.3	4.7	12.74	58.62
IND	3.11	60.33	5.59	5.77	6.38	10.77	6.36	1.7	39.67
UK	9.61	4.14	49.22	13.99	5.52	4.62	9.56	3.34	50.78
EUR	21.8	3.46	11.08	39.04	5.43	5.6	7.03	6.57	60.96
CHIN	6.17	5.54	6.52	7.59	55.16	9.39	6.05	3.57	44.84
KOR	4.29	9.14	5.46	7.56	9.32	53.4	8.38	2.46	46.6
CAN	5.59	5.34	10.21	8.94	5.72	8.44	53.98	1.79	46.02
JPY	18.02	1.34	4.11	9.86	3.75	2.13	1.8	58.99	41.01
Cont TO others	68.59	31.01	51.15	76.8	40.67	44.25	43.87	32.16	388.51
Cont incl own	109.97	91.34	100.38	115.83	95.84	97.65	97.85	91.15	TCI
Net spillovers	9.97	-8.66	0.38	15.83	-4.16	-2.35	-2.15	-8.85	48.56
<i>Before COVID-19</i>									
SFR	44.39	1.29	6.93	23.36	4.62	2.28	3.37	13.77	55.61
IND	2.05	64.72	3.13	4.18	6.97	13.23	3.34	2.37	35.28
UK	9.07	2.46	56.75	15.25	5.1	2.77	5.36	3.24	43.25
EUR	21.89	2.34	11	41.44	6.49	5.09	5.57	6.17	58.56
CHIN	6.01	5.76	4.94	8.62	52.96	11.36	5.55	4.81	47.04
KOR	3.04	11.13	2.85	7	11.52	54.96	6.62	2.89	45.04
CAN	4.42	3.27	5.61	7.88	6.03	7.52	62.73	2.54	37.27
JPY	17.96	1.51	3.14	8.02	4.57	2.25	2.33	60.22	39.78
Cont TO others	64.44	27.75	37.6	74.3	45.3	44.49	32.15	35.79	361.83
Cont incl own	108.83	92.48	94.35	115.75	98.26	99.45	94.88	96.01	TCI
Net spillovers	8.83	-7.52	-5.65	15.75	-1.74	-0.55	-5.12	-3.99	45.23
<i>During COVID-19</i>									
SFR	36.09	4.26	9.78	21.4	5.28	5.44	6.98	10.76	63.91
IND	5.55	50.28	9.28	9.17	5.96	7.39	10.88	1.47	49.72
UK	9.9	6.92	38.34	12.08	6	8.15	14.75	3.87	61.66
EUR	20.33	6.26	11.14	34.59	4.79	6.9	9.45	6.54	65.41
CHIN	7.55	5.65	8.42	7.22	55.46	5.85	6.65	3.2	44.54
KOR	6.24	6.78	10.58	8.66	5.54	49.41	11.09	1.72	50.59
CAN	7.03	9.02	15.88	10.17	5.55	9.81	41.54	0.99	58.46
JPY	16.75	2.38	6.21	10.94	3.84	2.64	1.84	55.41	44.59
Cont TO others	73.34	41.26	71.3	79.64	36.96	46.18	61.64	28.56	438.88
Cont incl own	109.43	91.54	109.64	114.23	92.41	95.59	103.19	83.97	cTCI/TCI
Net spillovers	9.43	-8.46	9.64	14.23	-7.59	-4.41	3.19	-16.03	62.70/54.86

Notes: This table reports the results of the average connectedness measures of cryptocurrencies based on Diebold and Yilmaz (2012, 2014) estimated from a TVP-VAR model for the full sample before and during the COVID-19 period. We provide the average of total connectedness, the directional spillover received (denoted by “From”), and transmitted (denoted by “Contribution to others”) by each variable. Then, the net directional spillover (denoted by “Net Spillovers”) is obtained as the difference between directional ‘To’ spillovers and directional ‘From’ spillovers.

connectedness among currencies. The quantile regression is specified via the following regression:

$$conn_t^{\tau} = \beta_0^{\tau} + \beta_1^{\tau} EPU_t + \beta_x^{\tau} X_t + \varepsilon_t^{\tau} \quad (9)$$

where $conn_t^{\tau}$ refers to τ^{th} the quantile of the total connectedness or the net connectedness index. EPU_t is the proxy for economic policy uncertainty. X_t represents the two other risk factors, namely, the CBOE volatility index (VIX) and the crude oil volatility index (OVX), and ε_t^{τ} is the error term. τ^{th} refers to the quantile order. If the coefficients on these factors, β_1^{τ} and β_x^{τ} , are statistically significant, this indicates that those factors significantly affect the dynamic spillover among currencies. This analysis gives more inclusive evidence of the relationship between uncertainty factors and the dynamic connectedness measures at different levels of the spillovers distribution.

Table 6 presents the estimated parameters and their corresponding probabilities over the full sample period, while Tables 7 and 8 present the results before and during the COVID-19 health crisis, respectively. Then, Figs. 6–8 depict the impact of EPU, VIX, and OVX on the total connectedness among all currencies across different quantiles over the full and the two sub-periods, respectively.⁵ As can be observed from these tables and figures, some estimated parameters are statistically significant for some quantiles, indicating that the connectedness between currencies is strongly influenced by these three factors. The results also confirm the presence of a heterogeneous impact of uncertainty on the dynamic connectedness for both types of currencies under different spillover distribution levels (e. g., low, normal, and high). Specifically, the results show a negative effect of the EPU on the total spillover among all currencies at all quantiles for both the full sample and COVID-19 periods. Therefore, an increase in uncertainty levels leads to a decrease in the level of connectedness among currencies (either traditional or crypto). These results can be explained by the reaction of investors to the increase in uncertainty. Investors may interpret uncertainty as bad news, which tends to increase their investment in currencies.

⁵ To save space, we only report the figures for the total connectedness and not those related to net connectedness of each cryptocurrency/traditional currency. Yet, they are available upon request.

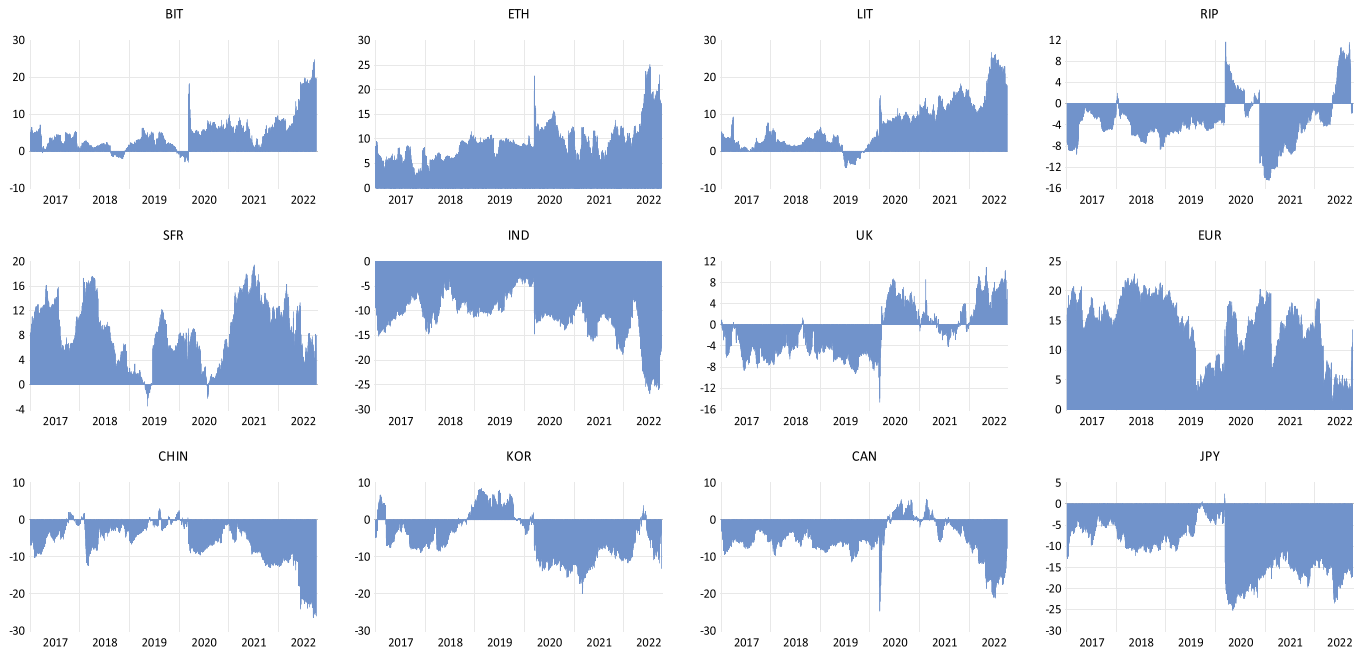


Fig. 3. : Dynamic net connectedness over the full sample period.

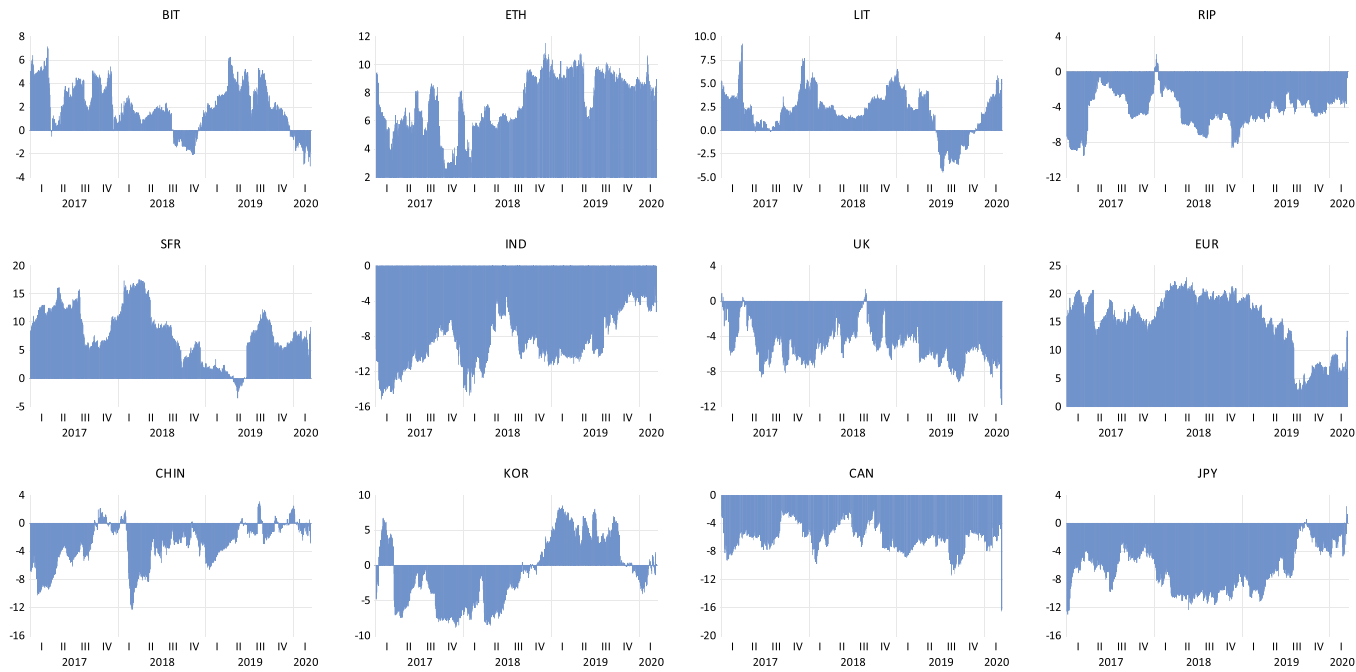


Fig. 4. : Dynamic net connectedness before the COVID-19 pandemic.

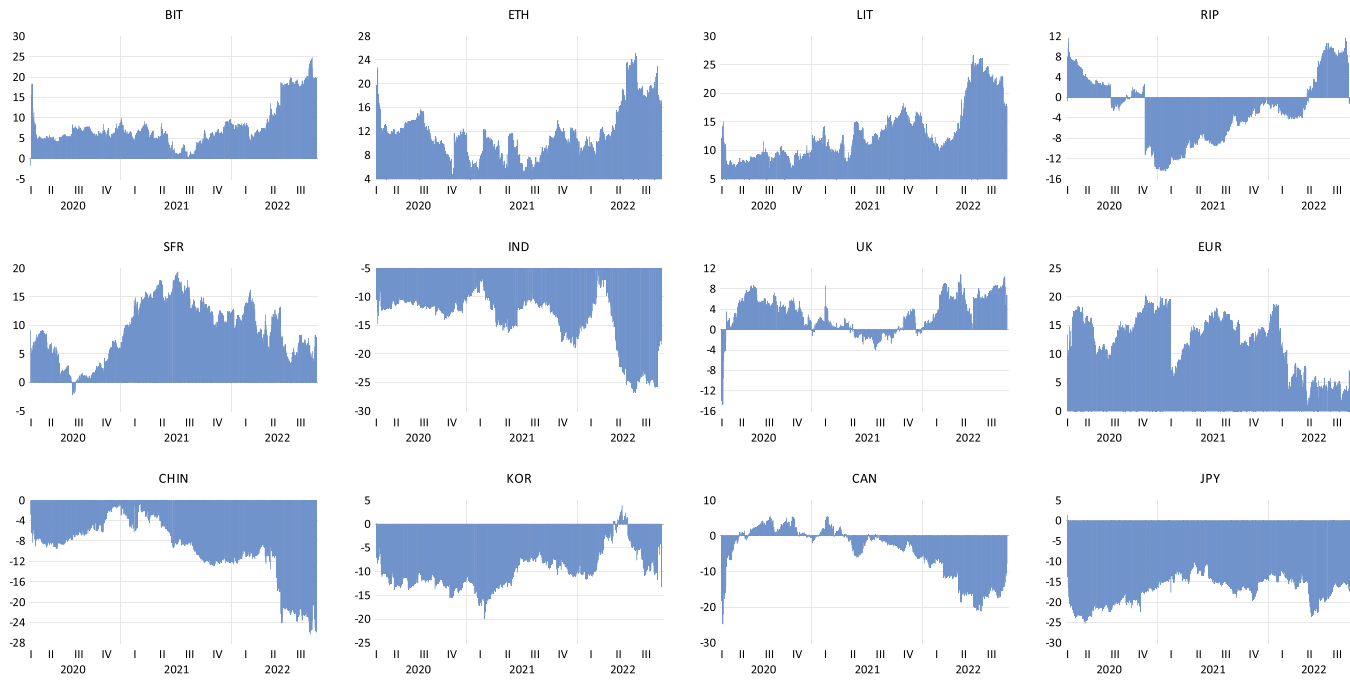


Fig. 5. : Dynamic net connectedness during the COVID-19 pandemic.

Table 6

Effect of EPU VIX and OVX on the total and net connectedness among currencies over the full sample period.

	EPU			VIX			OVX		
	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$
TOTAL	0.025 *** (0.000)	0.015 *** (0.000)	0.003 * (0.087)	0.345 *** (0.000)	0.411 *** (0.000)	0.020 *** (0.000)	-0.086 *** (0.0000)	0.0111 (0.454)	0.018 (0.735)
NET BIT	0.008 *** (0.000)	0.007 *** (0.000)	-0.003 ** (0.022)	-0.048 *** (0.001)	0.028 (0.363)	0.277 *** (0.000)	-0.008 * (0.037)	0.047 ** (0.027)	0.041 *** (0.000)
NET ETH	0.003 *** (0.000)	0.006 *** (0.000)	0.001 *** (0.000)	0.009 *** (0.000)	0.006 *** (0.000)	0.001 (0.385)	-0.004 (0.604)	0.032 * (0.076)	0.031 *** (0.000)
NET LIT	0.011 *** (0.000)	0.009 (0.000)	-0.003 (0.217)	0.195 *** (0.000)	0.284 *** (0.000)	0.571 * (0.056)	-0.055 *** (0.000)	0.015 (0.385)	0.207 (322)
NET RIP	-0.016 *** (0.000)	0.003 *** (0.000)	0.004 *** (0.000)	0.061 *** (0.000)	0.069 *** (0.000)	0.113 *** (0.000)	0.013 *** (0.000)	0.013 (0.327)	0.141 *** (0.000)
NET SFR	-0.005 *** (0.000)	-0.009 *** (0.000)	0.002 (0.344)	0.053 *** (0.000)	-0.056 * (0.010)	0.001 (0.995)	0.011 * (0.021)	0.000 (0.902)	-0.055 *** (0.000)
NET IND	0.007 *** (0.000)	-0.001 * (0.025)	-0.008 *** (0.000)	-0.371 (0.227)	-0.094 *** (0.000)	-0.050 (0.363)	0.002 (0.991)	0.025 (0.338)	0.033 (0.136)
NET UK	0.016 *** (0.000)	0.014 *** (0.000)	0.004 *** (0.000)	0.115 *** (0.000)	0.388 *** (0.000)	0.617 *** (0.000)	-0.057 *** (0.000)	-0.062 *** (0.000)	-0.066 * (0.034)
NET EUR	0.006 *** (0.000)	-0.006 *** (0.000)	-0.006 *** (0.000)	0.032 (0.814)	0.007 (0.850)	0.032 (0.267)	-0.152 (0.214)	-0.075 *** (0.000)	-0.059 *** (0.000)
NET CHIN	0.008 * (0.021)	0.001 (0.405)	-0.005 *** (0.000)	-0.364 * (0.090)	-0.299 *** (0.000)	-0.029 *** (0.017)	-0.001 (0.710)	0.034 * (0.023)	0.005 (0.248)
NET KOR	-0.022 *** (0.000)	-0.021 *** (0.000)	-0.023 *** (0.000)	-0.060 (0.189)	-0.104 *** (0.000)	-0.323 *** (0.000)	-0.029 (0.283)	0.044 *** (0.000)	0.075 *** (0.000)
NET CAN	0.021 *** (0.000)	0.02 *** (0.000)	0.023 *** (0.000)	-0.184 * (0.086)	-0.013 (0.646)	0.002 (0.949)	-0.146 * (0.024)	-0.011 (0.381)	-0.006 (0.228)
NET JPY	-0.011 *** (0.000)	-0.019 *** (0.000)	-0.021 *** (0.000)	-0.425 *** (0.000)	-0.429 *** (0.000)	-0.272 *** (0.000)	-0.134 * (0.071)	0.020 (0.357)	0.131 *** (0.000)

Notes: This table reports the estimated coefficients of the effect of EPU, VIX and OVX on the total and net dynamic connectedness, as in Eq. (9). The numbers between parentheses are the probabilities of the t-statistics. (***), (**), (*) indicate the parameter significance at 1%, 5%, and 10% significance level, respectively.

Table 7

Effect of EPU VIX and OVX on the total and net connectedness among currencies before the COVID-19 pandemic.

	EPU			VIX			OVX		
	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$
TOTAL	0.027 *** (0.000)	0.022 *** (0.000)	0.014 *** (0.000)	0.146 * (0.034)	0.199 * (0.029)	0.161 (0.474)	-0.022 (0.486)	-0.020 (0.233)	0.103 (0.363)
NET BIT	0.009 *** (0.000)	0.010 *** (0.000)	0.007 *** (0.000)	-0.065 *** (0.000)	-0.086 *** (0.000)	-0.119 *** (0.000)	-0.002 (0.212)	0.001 (0.916)	0.093 *** (0.000)
NET ETH	0.011 *** (0.000)	0.009 *** (0.000)	0.009 *** (0.000)	0.068 *** (0.000)	0.094 * (0.087)	0.036 (0.286)	0.001 (0.597)	0.018 (0.744)	0.073 *** (0.000)
NET LIT	0.011 *** (0.000)	0.011 *** (0.000)	0.010 (0.000)	0.142 * (0.062)	0.152 *** (0.000)	0.061 *** (0.002)	-0.034 (0.330)	-0.011 *** (0.002)	0.049 *** (0.000)
NET RIP	0.010 *** (0.000)	0.012 *** (0.000)	0.006 *** (0.000)	0.058 *** (0.000)	0.042 * (0.073)	0.075 *** (0.000)	0.002 (0.339)	0.001 (0.909)	0.131 *** (0.000)
NET SFR	-0.006 *** (0.000)	-0.014 *** (0.000)	-0.018 *** (0.000)	0.046 * (0.093)	-0.004 *** (0.000)	0.075 * (0.014)	0.011 * (-0.088)	-0.007 (0.411)	-0.055 *** (0.000)
NET IND	-0.001 (0.452)	-0.002 *** (0.000)	-0.009 *** (0.000)	0.059 *** (0.000)	-0.071 * (0.019)	0.005 (0.702)	-0.001 (0.822)	0.042 *** (0.000)	0.012 *** (0.000)
NET UK	0.011 * (0.071)	0.022 *** (0.000)	0.016 (0.000)	0.079 *** (0.000)	0.088 *** (0.000)	0.289 * (0.011)	-0.051 *** (0.000)	-0.045 *** (0.000)	-0.044 (0.409)
NET EUR	0.002 (0.297)	-0.012 *** (0.000)	-0.011 *** (-0.35)	0.197 (0.167)	0.186 *** (0.000)	0.064 * (0.051)	-0.103 (0.176)	-0.121 *** (0.000)	-0.062 *** (0.000)
NET CHIN	-0.000 (0.624)	-0.009 *** (0.000)	-0.010 *** (0.000)	-0.079 * (0.016)	0.014 (0.382)	-0.030 * (0.021)	0.024 *** (0.000)	0.007 (0.155)	0.007 (0.187)
NET KOR	-0.011 *** (0.000)	-0.022 *** (0.000)	-0.017 (0.212)	-0.202 (0.102)	-0.053 (0.101)	-0.049 (0.845)	-0.117 * (0.066)	0.027 * (0.029)	0.233 (0.119)
NET CAN	0.000 (0.998)	0.015 *** (0.000)	0.015 *** (0.000)	-0.155 * (0.030)	-0.186 *** (0.000)	-0.092 *** (0.000)	-0.191 *** (0.000)	-0.105 *** (0.000)	-0.009 *** (0.000)
NET JPY	-0.012 *** (0.000)	-0.023 *** (0.000)	-0.020 *** (0.000)	-0.594 *** (0.000)	0.406 *** (0.000)	-0.100 * (0.059)	-0.064 (0.360)	0.042 (0.230)	0.053 * (0.037)

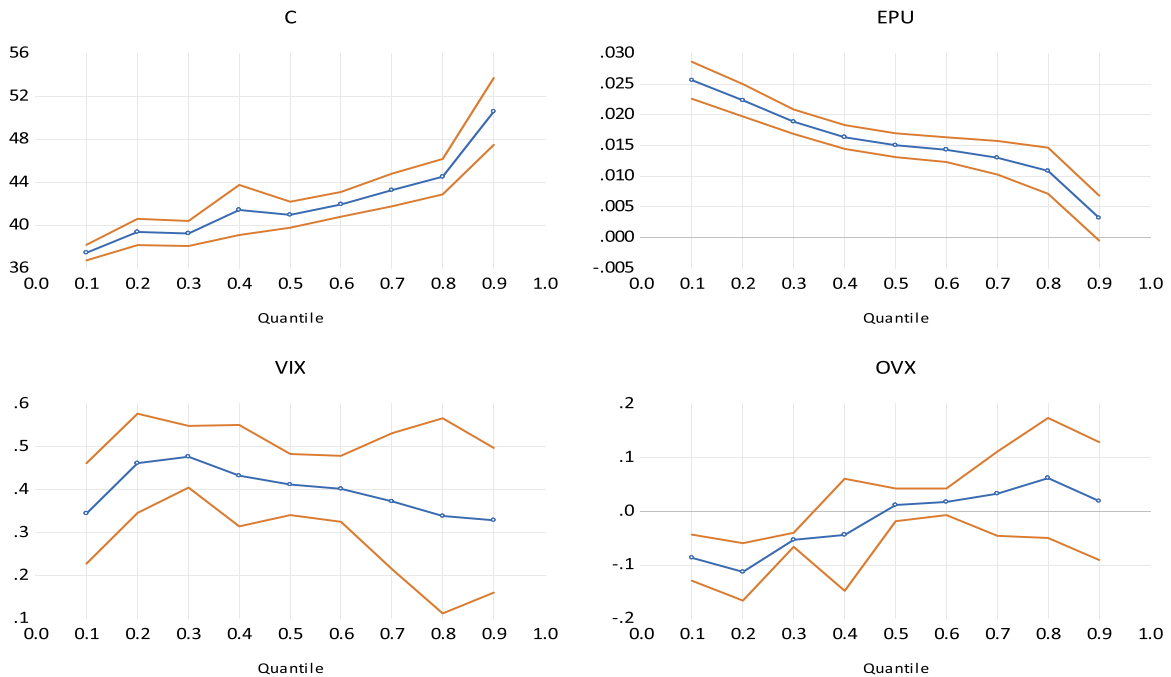
Notes: This table reports the estimated coefficients of the effect of EPU, VIX and OVX on the total and net dynamic connectedness, as in Eq. (9). The numbers between parentheses are the probabilities of the t-statistics. (***), (**), (*) indicate the parameter significance at 1%, 5%, and 10% significance level, respectively.

Table 8

Effect of EPU, VIX and OVX on the total and net connectedness among currencies during the COVID-19 pandemic.

	EPU			VIX			OVX		
	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$
TOTAL	0.027 *** (0.000)	0.022 *** (0.000)	0.017 *** (0.000)	0.014 ** (0.034)	0.199 ** (0.029)	0.161 (0.474)	-0.022 (0.485)	-0.020 (0.233)	0.103 (0.363)
NET BIT	0.009 *** (0.000)	0.010 *** (0.000)	0.007 *** (0.000)	-0.065 *** (0.000)	-0.086 *** (0.000)	0.119 *** (0.000)	-0.002 (0.212)	0.001 (0.916)	0.093 *** (0.000)
NET ETH	0.011 *** (0.000)	0.009 *** (0.000)	0.009 *** (0.000)	0.069 *** (0.000)	0.096 * (0.087)	0.036 (0.286)	0.001 (0.597)	0.018 (0.744)	0.073 *** (0.000)
NET LIT	0.011 *** (0.000)	0.011 *** (0.000)	0.010 *** (0.000)	0.142 * (0.062)	0.152 *** (0.000)	0.061 *** (0.000)	-0.034 (0.330)	-0.011 *** (0.000)	0.049 *** (0.000)
NET RIP	0.010 *** (0.000)	0.012 *** (0.000)	0.006 *** (0.000)	0.058 *** (0.000)	0.042 * (0.073)	0.076 *** (0.000)	0.002 (0.339)	0.000 (0.903)	0.131 *** (0.000)
NET SFR	-0.046 (0.000)	-0.014 (0.000)	-0.018 (0.000)	0.046 * (0.093)	-0.004 (0.856)	0.075 * (0.014)	0.011 * (0.088)	-0.007 (0.411)	-0.055 *** (0.000)
NET IND	-0.00079 (0.452)	-0.002 *** (0.000)	-0.009 *** (0.000)	0.059 *** (0.000)	-0.071 * (0.019)	0.005 (0.702)	-0.001 (0.822)	0.042 *** (0.000)	0.012 *** (0.000)
NET UK	0.011 * (0.071)	0.022 *** (0.000)	0.016 *** (0.000)	0.079 *** (0.000)	0.088 *** (0.000)	0.289 *** (0.011)	-0.0051 *** (4.45)	-0.045 *** (0.000)	-0.044 (0.409)
NET EUR	0.002 (0.297)	-0.012 *** (0.000)	-0.011 *** (0.000)	0.197 (0.167)	0.186 *** (0.000)	0.064 * (0.051)	-0.103 (0.176)	-0.121 *** (0.000)	-0.062 *** (0.000)
NET CHIN	-0.0002 (0.624)	0.009 *** (0.000)	0.010 *** (0.000)	-0.079 * (0.016)	0.014 (0.382)	-0.030 * (0.021)	0.024 *** (0.000)	0.007 (0.155)	0.007 (0.187)
NET KOR	-0.010 *** (0.000)	-0.021 *** (0.000)	-0.021 *** (0.000)	-0.177 *** (0.000)	-0.042 (0.177)	-0.174 ** (0.011)	0.009 (0.638)	0.027 * (0.028)	0.032 (0.336)
NET CAN	0.014 *** (0.000)	0.017 *** (0.000)	0.019 *** (0.000)	0.109 *** (0.000)	0.049 (0.110)	0.065 * (0.096)	-0.104 *** (0.000)	-0.023 * (0.081)	-0.014 * (0.018)
NET JPY	-0.012 *** (0.000)	-0.026 *** (0.000)	-0.0194 *** (0.000)	-0.561 *** (0.000)	-0.285 *** (0.000)	-0.099 (0.106)	-0.050 (0.555)	0.082 *** (0.000)	0.062 * (0.000)

Notes: This table reports the estimated coefficients of the effect of EPU, VIX and OVX on the total and net dynamic connectedness, as in Eq. (9). The numbers between parentheses are the probabilities of the t-statistics. (***), (**), (*) indicate the parameter significance at 1%, 5%, and 10% significance level, respectively.

**Fig. 6.** Quantile process estimates for the impact of EPU, VIX, and OVX on total connectedness for the full sample period.

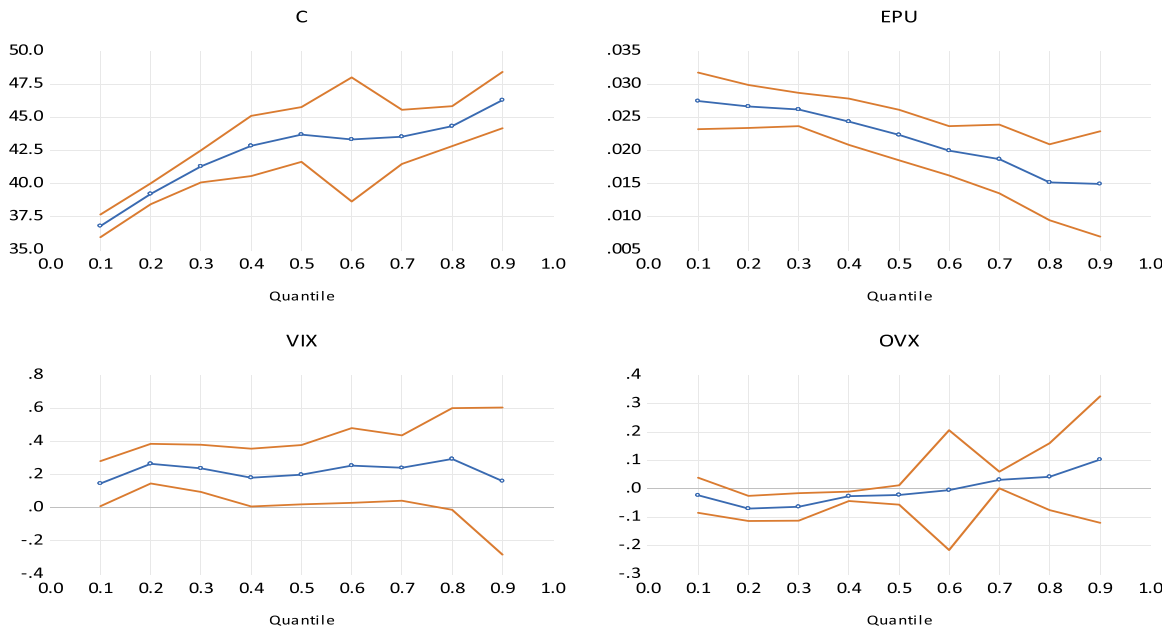


Fig. 7. Quantile process estimates for the impact of EPU, VIX, and OVX on total connectedness before COVID-19.

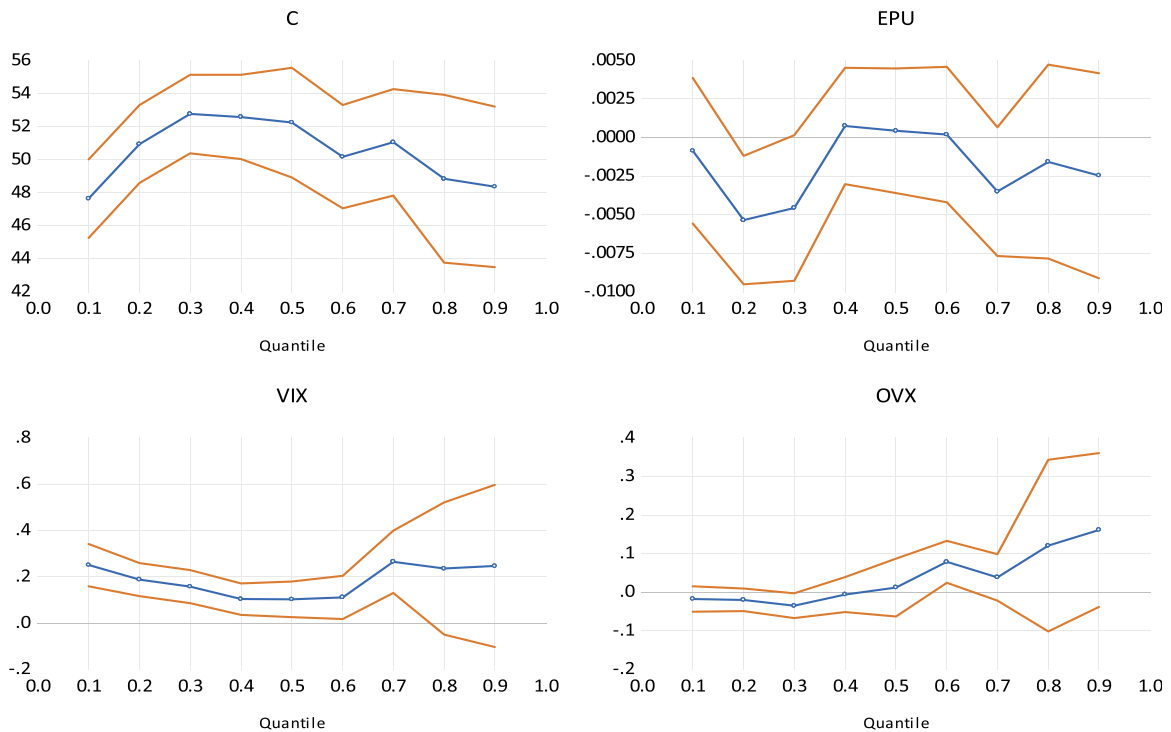


Fig. 8. Quantile process estimates for the impact of EPU, VIX, and OVX on total connectedness during COVID-19.

Regarding the impact of VIX and OVX, we observe the following: VIX positively impacts total connectedness at low quantiles and then decreases over high quantiles. This indicates that investors react positively to this risk factor and interpret that as good news under which they tend to increase their investment in these assets. The impact of OVX is negative at the low quantiles but then positive at high quantiles, indicating that the oil volatility exerts a negative effect on total connectedness when the market is in a stress condition, while total connectedness reacts positively when markets are at normal and high levels. The results of the total spillover before COVID-19 are similar to those of the full sample. We see that EPU has a significant negative impact on the total spillover among

all currencies at all quantiles. Although this impact fluctuates when the spillover effects are low, it shows a decreasing pattern when the level of connectedness is medium or high. The total spillover among all currencies becomes increasingly exposed to the negative impact of economic policy uncertainty in normal and high levels of connectedness.

Looking at the results of the net spillover in the full sample (as in Table 6), we notice more interesting results. The results report that EPU positively affects the net spillover of Bitcoin, Litecoin, and Ethereum at low quantiles but negatively for Ripple. Nevertheless, this impact turns negative at upper quantiles for Bitcoin and Litecoin and positive for Ethereum and Ripple. This means that the net spillover of Bitcoin and Litecoin become more exposed to the negative impact at the upper tail of the net connectedness distribution, making the two cryptocurrencies a good hedge against this market uncertainty in times of high-stress conditions.

Regarding the impact of VIX on the net connectedness, we notice a positive impact on the Swiss Franc, Indian Rube, The British pound, and the Euro at low quantiles, while there is a negative impact on the other traditional currencies. Those results also hold at the upper quantile. The negative impact of the fear index on some traditional currencies means that they are highly exposed to this factor, yet, they might play a good hedge against this market uncertainty at times of stress or when the markets are in times of crisis. Then in terms of the impact of OVX, we notice that most traditional currencies' net spillover reacts negatively to this factor at a low quantile level, except for the Swiss franc and Indian Rube. However, at a high quantile level of the distribution, oil volatility has an alternating impact. The Swiss franc, British pound, Euro, Canadian Dollar, and Japanese yen react negatively, while Indian Rube, Chinese Yuan, and Korean Won, react positively. This indicates again that some traditional currencies can play the role of diversifiers in times of stress or when the markets go through turmoil periods, such as those happening during the COVID-19 period.

Tables 7 and 8 again present the impact of the three factors on total connectedness and net spillover over the two sub-periods (before and during the COVID-19 crisis). Regarding the subperiod before COVID-19 period, Table 7 indicates that EPU positively affects the net spillover of the four cryptocurrencies at low and high quantile levels, meaning that the four assets react positively to this factor. However, the results are different for the impact of VIX on the four cryptocurrency net spillovers. Bitcoin is the only cryptocurrency that reacts negatively to the fear index under all market conditions, making it a good hedge against this market volatility under all market conditions. Also, Bitcoin reacts negatively to OVX at a low quantile level, along with Litecoin, implying the importance of these two cryptos for hedging and diversification benefits against the changes in fluctuations of oil prices during times of stress.

Before the COVID-19 pandemic, EPU had a significant positive effect on the net connectedness of the British Pound, and Canadian Dollar, at all quantiles. The EPU positively impacts the Euro in the lower quantile, yet, turns negative at the medium and high quantiles. Swiss France, Indian Rube, Chinese Yuan, Korean Won, and Japanese Yen all react negatively to the EPU under all market conditions. Regarding the fear index (VIX), it is noticed that British Pound, and the Euro, react positively to this factor under all market conditions, while the rest alternate in terms of their reaction, depending on whether the market is in a bearish or bullish condition. For example, the Swiss franc is negatively reacting to VIX at low and high quantiles, making this currency a good hedge against this market uncertainty under extreme market stress. The same applied to Indian Rube, Korean Won, Canadian Dollar, and Japanese Yen. When it comes to OVX, we notice a negative reaction of the British Pound, The Euro, and the Canadian Dollar to the oil risk factor under all market conditions, while the Swiss Franc is affected negatively when the market is in a bullish state.

The results during the COVID-19 period are surprisingly different for traditional currencies. Although some of the traditional currencies (Swiss franc, Indian Rupee, Korean Won, and the Japanese Yen) are exposed to a negative impact of EPU, the net spillovers for other traditional currency show evidence of a positive relationship with EPU (e.g., the Pound, and the Canadian Dollar). For the impact of VIX on the traditional currencies' net spillover, Korean Won and Japanese Yen react negatively at all quantiles, while the British Pound, the Euro, and the Canadian Dollar react positively at all quantiles. That makes the Korean Won and Japanese Yen good hedges against this market uncertainty during the COVID-19 period under all market conditions. The Chinese Yuan also reacts negatively to VIX when the market is in bearish and bullish states, making this currency another good candidate for hedging the fear market uncertainty under extreme market conditions. The Swiss Franc and Indian Rube are affected positively under extreme market conditions. When it comes to the oil price volatility (OVX), the British Pound, the Euro, and the Canadian Dollar, are all affected negatively by this factor, providing investors with some hedging opportunities during the COVID-19 period against this market uncertainty. The Indian Rube, Chinese Yuan, Korean Won, and Japanese Yen, react positively to oil price volatility under all market conditions. The Swiss Franc also is negatively impacted at a high quantile, making it a good candidate for hedging this uncertainty during the COVID-19 period.

Our results fall in-line with those by Demir et al. (2018), Dyhrberg (2016), and Briere et al. (2015), among others. For example, Demir et al. (2018) suggested that Bitcoin is an effective tool for hedging, especially during bullish market conditions, while it showed some diversification benefits during bearish market conditions. Dyhrberg (2016) further suggested that Bitcoin can be an advantageous instrument for portfolio and risk management when it is isolated from other financial assets. Briere et al. (2015), Corbet et al., (2018, 2021), and Lee et al. (2018) also reported that Bitcoin offers diversification benefits due to being weakly correlated with other traditional assets. Yet, the disadvantage of cryptocurrency, as explained by Kristoufek (2018), was the low liquidity of the cryptocurrency market compared to other financial markets, like the equity and foreign exchange markets.

5. Conclusion

The paper examined the dynamic connectedness among eight traditional currencies and four cryptocurrencies before and over the COVID-19 pandemic periods. We found strong evidence of dynamic spillover across currency markets, with a noticeable increase during the COVID-19 pandemic period. During the COVID-19 period, a positive jump in the total spillover of both types of currencies was clearly noticed, reaching its highest level between the end of 2020 and the beginning of 2021. Then, it fell somehow during 2021

but then increased during the 2022 period, marking the impact of the Russian-Ukraine war. The fall in total spillover in the early days of the crisis indicates that investors had less fear and became more confident because of the final confirmation of the COVID-19 vaccination discovery.

Our results revealed that the total spillover among traditional currencies is slightly higher than the spillover among cryptocurrencies during the crisis periods. Before COVID-19, cryptocurrencies were found to be the most transmitters of shocks compared to traditional currencies, with Ethereum being the highest transmitter and Ripple the lowest one. The highest traditional currencies transmitters are the Swiss Franc and the Euro, while the lowest transmitters are the Indian Rupee and the Japanese Yen. We also reported evidence of a significant rise in the level of volatility transmission across all types of currencies during the pandemic crisis, where the majority of traditional currencies and cryptocurrencies became the highest transmitters. These results confirm that the dominance of currencies in the shock transmission changes from one currency to another before and during COVID-19. Traditional currencies (EUR, UK, SFR, and CAN) as well as cryptocurrencies (BIT, ETH, and LIT) are the dominant transmitters of shocks during the crisis, with the Euro, British Pound, and Swiss Franc being the most interconnected with other currencies. In addition, Litecoin played an important role as a safe haven asset that dominated Bitcoin during the pandemic crisis. Overall, during the COVID-19 period, the dependence was found to be positive and higher for the pairs of the same market than those across markets. As both markets (cryptocurrencies and fiat currencies) are weakly connected, investors might benefit more from holding these assets together for portfolio diversification purposes.

Looking at the two markets as separate groups, total spillover among cryptocurrencies, on average, during the COVID-19 crisis was found to be higher than before the crisis period. Ethereum and Litecoin are the highest contributors to other cryptocurrencies during and before the crisis. That shows cryptocurrencies to be considered as good diversifiers for portfolio construction during the pandemic period, with a noticeable increase in contribution levels during the pandemic period. The net spillover among cryptocurrencies indicates that Ethereum is the highest net transmitter before the crisis, while Ripple is found to be the highest net receiver. Overall, during the COVID-19 period, cryptocurrencies showed evidence of more net transmitters and net receivers than before COVID-19.

Similarly, the results obtained for the traditional currencies showed that the total spillover among traditional currencies, on average, during the COVID-19 crisis, is higher than their total spillover before crisis period. The highest contributors to other traditional currencies are the Swiss Franc, Euro, Chinese Yuan, Korean Won, and Canadian Dollar in both sub-periods. Yet, during the crisis, there is a noticeable increase in the level of contribution of the traditional currencies compared to the period before the pandemic. Before the crisis, two traditional currencies were net transmitters (the Swiss Franc and the Euro), while the others were net receivers. The highest net transmitter is the Euro, whereas the Indian Rupee is the highest net receiver. However, the results of net spillover are somewhat different during the COVID-19 period. The net transmitters are the Swiss Franc, the British Pound, the Euro, and the Canadian Dollar, with the Euro being the highest transmitter. The position of British Pound and Canadian Dollar is changed from being net receivers before the crisis to net transmitters during the crisis. The other currencies maintained their positions as net receivers of shocks during the pandemic period. The highest net receivers are the Indian Rupee and the Japanese Yen. Overall, traditional currencies display a higher level of net spillover during the COVID-19 period.

The paper also provided evidence on the impact of economic policy uncertainty (EPU) on total spillovers and net spillovers among cryptocurrencies and traditional currencies. Our results confirmed the presence of a heterogeneous impact of EPU on the dynamic connectedness for both types of currencies under different spillover distribution levels (e.g., low, normal, and high). Specifically, the results show a negative effect of EPU on the total spillover among all currencies at all quantiles for both the full sample and COVID-19 periods. These results can be explained by the reaction of investors to the increase in uncertainty. Investors may interpret uncertainty as bad news, which tends to increase their investment in currencies. In general, we found that EPU has a significant negative impact on the total spillover among all currencies at all quantiles. Although this impact fluctuates when the spillover effects are low, it shows a decreasing pattern when the level of connectedness is medium or high. The total spillover among all currencies becomes increasingly exposed to the negative impact of EPU in normal and high levels of connectedness.

Regarding the impact of VIX and OVX on the level of connectedness, contradictory results were reported. VIX positively impacted total connectedness at low quantiles, then decreased gradually over the high quantiles. This indicates that investors react positively to this risk factor and interpret that as good news under which they tend to increase their investment in these assets. However, OVX showed a negative impact on the total connectedness at low quantiles but then turned positive at high quantiles. This means that oil market volatility exerts a negative effect on total connectedness when the market is in a stressful condition. In contrast, total connectedness reacts positively to OVX when markets are at normal and high levels.

Our findings have many important policy implications. First, it can help investors select the appropriate currencies that lead to better portfolio diversification. Indeed, our results suggest that investors in the foreign exchange market can use cryptocurrency investment as a hedge against the risk associated with the foreign exchange market. Second, the nature of the relationship between economic policy uncertainty and the connectedness of different currencies with cryptocurrencies can offer useful information to promote economic prosperity, especially in periods of extreme market conditions, such as the COVID-19 pandemic. Third, our paper is also of great significance to policymakers. They can rely on our results to properly manage foreign currency risk and other potential risks associated with currency markets. This way, policymakers can control the risks in currencies of significant connectedness and then decide on the timing and extent of foreign currency rate policy intervention. Finally, unveiling the causes of currency price variations leads to a better selection of the source of oscillation that affects economic stability differently.

This study could be extended in the future in some ways. First, the sample can be extended to include other cryptocurrencies or traditional currencies to examine their connectedness across the virtual and foreign exchange markets. Second, it can also include other asset classes, such as the recently created NFTs or other virtual assets, to explore their hedging and safe haven properties for cryptocurrencies under different market conditions and their response to policy uncertainty indices. Third, different research

methodologies can be used, like copula or multivariate dependence models, to study the dependence structure. Fourth, analyzing the risk benefits of combining the two markets in terms of their risk value, expected shortfall, or other risk measures during normal and crisis periods can be used as another aspect for future research. Finally, applying a tail-dependence approach within the Diebold-Yilmaz framework to analyze the connectedness among cryptos, traditional assets, and other alternative assets might be a fruitful line for future research.

Declaration of interests

The authors declare that they have no financial and personal relationships with other people or organizations that could inappropriately influence (bias) their work.

Data Availability

Data will be made available on request.

Appendix A. Literature review summary

References	Targeted assets	The model	Global crises	Findings
Volatility connectedness among global currency markets				
Kitamura (2010)	AUD, GBP, CAD, EUR, JPY, CHF	M-GARCH model	The 2008 global financial crisis (GFC)	Volatility connectedness from the Euro significantly impacts the Swiss franc and Japanese yen during the global financial crisis (GFC) period.
Baruník et al. (2017)	AUD, GBP, CAD, EUR, JPY, CHF	2 N-dimensional VAR model		Negative volatility connectedness among currencies dominated positive ones.
Greenwood-Nimmo et al. (2016)	G10 currencies	Empirical network model	The 2008 global financial crisis (GFC)	Volatility connectedness among currencies was time-varying and it increased widely during financial crises.
Salisu and Ayinde (2018)	USD, EUR, GBP, JPY, CHF, the West African Unit of Account (WAUA)	The VAR model	Around the election date in Nigeria and the GFC	Volatility connectedness among Naira and six most-traded currencies rose during the election process in Nigeria more than during the GFC.
Mai et al. (2018)	Global currency markets, East Asian and the European currency markets	Currency correlation matrix and impact elimination method		The East Asian currencies were more strongly connected than the European currencies due to the strong co-movement of currencies in the East Asian region.
Antonakakis (2012)	Major foreign currency rates in Europe	Dynamic correlations and VAR-based spillover	Before and after the introduction of the Euro currency	Volatility connectedness between major foreign currency rates was weak in the post-euro period.
Kočenda and Moravcová (2019)	Three currencies in the new EU foreign exchange markets	Dynamic Conditional Correlation (DCC) model and the generalized vector autoregressive (GVAR) variance decomposition	Before the GFC and during GFC as well as during the European debt crisis	Volatility connectedness among currencies rose significantly rose during the GFC.
Return connectedness among global currency markets				
Shu et al. (2015)	The Chinese Yuan and Asia-Pacific countries' currencies	Multiple regression with causality	Before and after the transmission of China's monetary stance	The offshore currency exchange of the Chinese currency exerted more effect on the Asia-Pacific currencies than the onshore Chinese currency market due to China's monetary policy transmission.
Sehgal et al. (2017)	South East Asian currencies	Constant and time-varying Copula-GARCH models		Connectedness among currencies return of the South Asian member countries was very weak.
Wei et al. (2020)	The Belt and Road currency market	Time-varying parameter vector autoregressive (TVP-VAR) framework	During the regional and global crises including COVID-19	Connectedness among RMB and "the Belt and Road" currencies was stronger during the regional and the GFC crises but it was disrupted during the COVID-19 period due to internal financial reforms, as well as external economic shocks in the region.

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References	Targeted assets	The model	Global crises	Findings
Orlowski (2016)	The Euro and the non-Euro currencies	GARCH and BVAR models,	During the GFC	The positive connectedness between the Euro and the non-Euro currencies rose significantly during the GFC.
McCauley and Shu (2019)	The Chinese currency (RMB) with regional and other emerging market currencies	Simple and multiple regressions	After the three post-reform periods of RMB management (transition, basket of currencies management, and countercyclical management)	Connectedness between regional and Latin American currencies rose in the basket period but declined after the countercyclical period.
Volatility connectedness among cryptocurrencies and/or traditional currencies				
Kumar and Anandaramo (2019)	Bitcoin, Ethereum, Ripple, and Litecoin	Multivariate IGARCH-DCC and the pairwise wavelet cross-spectral analysis		Volatility connectedness was very strong Bitcoin and other cryptos (Ethereum and Litecoin).
Wen and Wang (2020)	65 major global currencies	LASSO-VAR approach	Oil price crashes, exchange rate regimes, and monetary policy changes)	The sensitivity between volatility connectedness among the US dollar and the Euro and changes in international economic fundamentals increased during crisis periods (oil price crashes, exchange rate regimes, and monetary policy changes). Volatility connectedness among cryptocurrencies persist over time.
Fung et al. (2022)	254 cryptocurrencies	GARCH family models		
Return connectedness among traditional currencies and/or cryptocurrencies				
Baumöhl (2019)	Six traditional currencies, and six cryptocurrencies	The quantile cross-spectral approach		A negative connectedness was found between currencies and cryptocurrencies in both the short- and long-term horizons.
Hsu et al. (2021)	Cryptocurrency and traditional currencies or gold markets	Using a diagonal BEKK model	During COVID-19	Cryptocurrencies were strongly connected with traditional currencies and gold, acted as a safe haven during the COVID-19 pandemic against risk spillover.
Mokni and Ajmi (2020)	Bitcoin, Ethereum, Ripple, and Litecoin and Bitcoin Cash and the US dollar	Granger-causality in quantiles	Before and during the COVID-19 period	During the COVID-19 crisis, the connectedness between the US dollar and cryptocurrencies was stronger at higher and lower tails of the distribution and cryptocurrencies were good predictors, and acted as hedgers against the US dollar volatility.
Elsayed et al. (2022)	Bitcoin, Litecoin, Ripple, and nine major foreign currency markets	The generalized vector autoregressive (GVAR) variance decomposition and the Bayesian graphical structural vector autoregressive estimations	During the 2017–2018 cryptocurrency crash	Bitcoin and Litecoin were the most connected currencies during the three quarters of 2017. Except for the Chinese Yuan, major traditional currencies did not significantly impact cryptocurrencies.
Bouri et al. (2021)	Bitcoin, Ethereum, Ripple, Litecoin, Stellar, Monero, Dash	Quantile VAR approach	During COVID-19	The connectedness among cryptocurrencies rose widely during the COVID-19 outbreak. The level of connectedness increases with increase in shock size either for both positive or negative shocks.
Kumar et al. (2022)	Bitcoin, Ethereum, Ripple, Litecoin, Bitcoin Cash, EOS, Binance Coin, Tether, Bitcoin SV, and Tron	The generalized vector autoregression (VAR) framework and Granger causality	During COVID-19	The structural change resulted from the COVID-19-pandemic impacted the connectedness among cryptocurrencies.
EPU and connectedness among traditional currencies and/or among cryptocurrencies				
Mokni et al. (2020)	EPU and connectedness between Bitcoin and the US stock markets	DCC-EGARCH model and simple regression	After the crash of Bitcoin in December 2017	EPU negatively affected the connectedness among Bitcoin and the US stock markets only after the crash of Bitcoin in December 2017, suggesting that Bitcoin as a hedging instrument to the stock portfolio.
Huynh et al. (2020)	TPU and connectedness between the US exchange rates of globally traded currencies	The generalized vector autoregressive (VAR) variance decomposition		Trade policy uncertainty impacted the connectedness among the US dollar exchange rates of globally traded currencies.

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References	Targeted assets	The model	Global crises	Findings
Chen et al. (2020)	EPU and China's exchange rate volatility	The quantile regression approach		EPU positively affected China's exchange rate volatility on all quantiles.
Alam et al. (2019)	Oil, foreign currency futures for AUD, CAD, CHF, EUR, GBP and JPY against the US dollar	Wavelet-Granger causality method of Olayeni (2016)	During the GFC period and the European sovereign debt crisis	Oil prices were strongly connected to exchange rates of six major bilateral currencies against the US dollar.
Al-Shboul et al. (2022)	Bitcoin, Ripple, Litecoin and Ethereum	Quantile VAR approach	During the COVID-19 pandemic	The effect of cryptocurrency uncertainty on the return connectedness increased significantly during the COVID-19 crisis.

Appendix B. Total average dynamic connectedness based on QVAR

Full Sample	BIT	ETH	LIT	RIP	SFR	IND	UK	EUR	CHIN	KOR	CAN	JPY	FROM
<i>q</i> = 0.05													
Cont TO others	84.51	87.35	85.53	81.80	92.23	82.69	90.43	94.31	82.97	88.11	87.16	85.52	1042.61
Cont incl own	97.34	99.97	98.67	95.41	105.08	95.57	103.42	107.32	96.50	101.30	100.39	99.01	TCI
Net spillovers	-2.66	-0.03	-1.33	-4.59	5.08	-4.43	3.42	7.32	-3.50	1.30	0.39	-0.99	86.88
<i>q</i> = 0.5													
Cont TO others	65.36	72.13	70.04	51.30	64.31	30.06	50.25	73.32	38.07	41.94	43.74	27.84	628.36
Cont incl own	105.19	110.26	108.43	97.91	106.43	88.94	97.75	112.22	94.63	95.22	95.00	88.02	TCI
Net spillovers	5.19	10.26	8.43	-2.09	6.43	-11.06	-2.25	12.22	-5.37	-4.78	-5.00	-11.98	52.36
<i>q</i> = 0.95													
Cont TO others	87.57	88.60	86.29	85.88	89.80	85.22	87.70	92.35	86.21	85.56	85.92	83.42	1044.53
Cont incl own	100.43	101.93	99.81	98.90	102.51	97.69	100.61	104.94	99.24	98.29	99.04	96.59	TCI
Net spillovers	0.43	1.93	-0.19	-1.10	2.51	-2.31	0.61	4.94	-0.76	-1.71	-0.96	-3.41	87.04
Before COVID-19													
<i>q</i> = 0.05													
Cont TO others	87.80	89.86	83.88	80.82	89.81	78.87	91.78	93.97	84.90	89.79	87.26	85.55	1044.30
Cont incl own	100.26	102.31	97.31	94.52	102.30	91.69	104.56	107.02	97.56	103.22	100.49	98.76	TCI
Net spillovers	0.26	2.31	-2.69	-5.48	2.30	-8.31	4.56	7.02	-2.44	3.22	0.49	-1.24	87.03
<i>q</i> = 0.5													
Cont TO others	56.59	65.93	57.79	45.83	59.30	27.11	35.03	70.47	43.90	45.21	32.90	28.15	568.20
Cont incl own	102.04	107.82	101.73	96.52	106.26	91.34	92.60	113.26	98.94	100.46	94.99	94.03	TCI
Net spillovers	2.04	7.82	1.73	-3.48	6.26	-8.66	-7.40	13.26	-1.06	0.46	-5.01	-5.97	47.35
<i>q</i> = 0.95													
Cont TO others	87.10	88.01	87.42	84.46	90.43	85.35	85.80	93.35	89.80	85.26	83.57	84.15	1044.68
Cont incl own	99.75	100.80	101.14	97.05	103.40	97.61	99.40	105.90	102.63	97.74	97.18	97.42	TCI
Net spillovers	-0.25	0.80	1.14	-2.95	3.40	-2.39	-0.60	5.90	2.63	-2.26	-2.82	-2.58	87.06
During COVID-19													
<i>q</i> = 0.05													
Cont TO others	85.68	85.22	89.17	83.43	93.74	83.31	89.07	92.27	80.23	88.18	87.24	82.46	1040.01
Cont incl own	99.23	98.44	102.43	97.31	106.77	95.47	102.45	105.22	94.56	101.49	100.44	96.18	TCI
Net spillovers	-0.77	-1.56	2.43	-2.69	6.77	-4.53	2.45	5.22	-5.44	1.49	0.44	-3.82	86.67
<i>q</i> = 0.5													
Cont TO others	71.46	75.07	81.15	50.18	71.34	33.77	62.54	75.39	29.57	42.51	51.00	27.97	671.95
Cont incl own	108.58	112.71	116.73	98.14	108.76	85.88	101.24	110.91	89.39	91.99	91.86	83.81	TCI
Net spillovers	8.58	12.71	16.73	-1.86	8.76	-14.12	1.24	10.91	-10.61	-8.01	-8.14	-16.19	56.00
<i>q</i> = 0.95													
Cont TO others	90.21	89.64	86.48	86.63	89.28	84.83	88.85	92.77	78.75	85.05	89.46	83.67	1045.63
Cont incl own	103.53	103.43	99.59	99.82	101.60	97.32	100.95	105.11	91.80	97.47	102.47	96.91	TCI
Net spillovers	3.53	3.43	-0.41	-0.18	1.60	-2.68	0.95	5.11	-8.20	-2.53	2.47	-3.09	87.14

Notes: This table presents the average of connectedness measures based on Diebold and Yilmaz (2012, 2014) estimated from a Q-VAR model. For each quantile's order, we provide the directional volatility spillover received (denoted by "From"), and transmitted (denoted by "Cont TO others") by each variable. Then, we obtain the net directional spillover (denoted by "Net Spillovers") as the difference between directional 'To' spillovers and directional 'From' spillovers.

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